

Evolutionary unique *N*-glycan-dependent protein quality control system plays pivotal roles in cellular fitness and extracellular vesicle transport in *Cryptococcus neoformans*

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This **important** study confirms the molecular function of putative components of the *N*-glycan-dependent endoplasmic reticulum protein quality control (ERQC) system in the pathogen *Cryptococcus neoformans*. The study demonstrates an involvement in fitness, virulence, and the secretion and composition of extracellular vesicles, albeit in ways that are not yet fully understood. The evidence provided is **convincing**, with rigorous, well-controlled assays and the use of complemented strains.

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Abstract

A conserved *N*-glycan-dependent endoplasmic reticulum protein quality control (ERQC) system has evolved in eukaryotes to ensure accuracy during glycoprotein folding. The human pathogen *Cryptococcus neoformans* possesses a unique *N*-glycosylation pathway that affects microbial physiology and interactions with the infected host. To investigate the molecular features and functions of the ERQC system in *C. neoformans*, we characterized a set of mutants with deletion of genes coding for the ERQC sensor UDP-glucose:glycoprotein glucosyltransferase (*UGG1*) and putative α 1,2-mannose-trimming enzymes (*MNS1*, *MNS101*, *MNL1*, and *MNL2*). The *ugg1* Δ , *mns1* Δ , *mns101* Δ , and *mns1* Δ 101 Δ mutants showed alterations in *N*-glycan profiles, defective cell surface organization, decreased survival in host cells, and varying degrees of reduced *in vivo* virulence. The *ugg1* Δ strain exhibited severely impaired extracellular secretion of capsular polysaccharides and virulence-related enzymes. Comparative transcriptome analysis showed the upregulation of protein folding, proteolysis,

and cell wall remodeling genes, indicative of induced ER stress. However, no apparent changes were observed in the expression of genes involved in protein secretion or capsule biosynthesis. Additionally, extracellular vesicle (EV) analysis combined with proteomic analysis showed significant alterations in the number, size distribution, and cargo composition of EVs in *ugg1Δ*. These findings highlight the essential role of the functional ERQC system for cellular fitness under adverse conditions and proper EV-mediated transport of virulence factors, which are crucial for the full fungal pathogenicity of *C. neoformans*.

Introduction

Glycoproteins, destined for the secretory pathways, enter the endoplasmic reticulum (ER) lumen where protein folding occurs (Rapoport, 2007). The accumulation of misfolded proteins affects cell viability and homeostasis; therefore, eukaryotes evolved a conserved ER quality control (ERQC) system that recognizes folding defects, repairs them, or ensures the translocation of irreparable misfolded proteins into the cytosol for proteasome-mediated degradation via the ER-associated degradation (ERAD) system (Thibault and Ng, 2012; Xu and Ng, 2015; Balchin et al., 2016), a process that heavily relies on *N*-glycosylation to determine protein folding conformation (Aebi, 2013; Varki, 2017).

Many aspects of *N*-glycosylation are highly conserved, and most eukaryotes initiate this process by synthesizing a Dol-PP-linked Glc₃Man₉GlcNAc₂ oligosaccharide as a common core *N*-glycan, which attaches to nascent polypeptides in the ER via an asparagine residue. Glucosidases I and II (Gls1 and Glc2) remove two glucose residues from the core oligosaccharide. Proteins containing monoglucosylated *N*-glycans (Glc₁Man₉GlcNAc₂) enter a calnexin and/or calreticulin (CNX/CRT in mammals; Cne in yeast) chaperone-mediated folding cycle. Finally, Glc2 cleaves the remaining glucose residue. If proteins are misfolded, they are recognized by the ERQC checkpoint enzyme UDP-glucose:glycoprotein glucosyltransferase (UGGT), which reglucosylates them for re-entry into the folding cycle (Fig. 1A). Then, the *N*-glycans of the accurately folded proteins are further processed by Glc2 and α1,2-mannosidase I (Mns1) and moved to the Golgi apparatus. However, irreparably misfolded glycoproteins are targeted for ERAD, where they undergo demannosylation and retro-translocation for proteasomal degradation in the cytosol. Recently, *N*-glycan precursors in the ER of some protists and fungi were found to be shorter than the typical 14-sugar *N*-glycan precursors in most eukaryote organisms. The length of these *N*-glycan precursors significantly impacts *N*-glycan-dependent QC of glycoprotein folding and ERAD (Banerjee et al., 2007; Samuelson et al., 2015).

The basidiomycetous fungus *Cryptococcus neoformans* is an opportunistic encapsulated human pathogen that primarily affects immunocompromised individuals, causing fatal meningoencephalitis (Gottfredsson and Perfect, 2000; Kwon-Chung et al., 2000). The *N*-glycosylation pathway of *C. neoformans* is evolutionarily conserved; nevertheless, the structure and biosynthesis of its *N*-glycans includes several unique features (Park et al., 2012). *C. neoformans* contains serotype-specific high-mannose-type *N*-glycans with or without a β-1,2-xylose residue attached to the trimannosyl core. Additionally, acidic *N*-glycans of *C. neoformans* contain xylose phosphates attached to the mannose residues both within the *N*-glycan core and outer mannose chains. The intact core *N*-glycan structure is crucial for *C. neoformans* pathogenicity (Thak et al., 2020); hence, alterations in the *N*-glycan structure modulates the interaction between the cell surface mannoproteins and host cells (Lee et al., 2023). Additionally, *C. neoformans* lacks homologous genes to the Asn-Linked Glycosylation (ALG) genes *ALG6*, *ALG8*, and *ALG10*, which are evolutionary conserved across most eukaryotic organisms and encode the glucosyltransferases that add the glucose residues to the core *N*-glycan before its attachment to proteins (Park et al., 2012). Man₇GlcNAc₂ and Man₈GlcNAc₂ without glucose residues are primarily detected in Dol-PP-linked glycans of *C. neoformans* (Samuelson et al., 2005). Moreover, the mature core *N*-glycan structures assembled on the cell surface mannoproteins of *C.*

C. neoformans are primarily $\text{Man}_6\text{--}7\text{GlcNAc}_2$, which are shorter than the expected $\text{Man}_8\text{GlcNAc}_2$ (Park et al., 2012 [DOI](#)). This observation led to the speculation that the terminal α -1,2-mannose residues of *C. neoformans* N-linked glycans may be more susceptible to trimming by ER α -1,2 mannosidases due to the lack of glucose residues compared to most eukaryotes. Alternatively, the presence of multiple α -1,2 mannosidases may generate more extensively trimmed core N-glycans in *C. neoformans*.

C. neoformans employs several virulence factors, including an extensive polysaccharide capsule composed of glucuronoxylomannan (GXM) and galactoxylomannan (GalXM) (Doering et al., 2009 [DOI](#)), melanin (Qiu et al., 2012 [DOI](#)), and various extracellular enzymes such as phosphatase and urease (Singh et al., 2013 [DOI](#)), all of which contribute to immune evasion and enhance fungal pathogenicity. To facilitate the export of these virulence-associated molecules, *C. neoformans* employs both conventional (Yoneda and Doering, 2006 [DOI](#); Panepinto et al., 2009 [DOI](#)) and unconventional secretion pathways (Rodrigues et al., 2007 [DOI](#); Casadevall et al., 2019 [DOI](#); Rizzo et al., 2021 [DOI](#)). Classical secretion relies on the ER-Golgi network, while non-classical mechanisms, such as extracellular vesicle (EV)-mediated transport, provide an alternative route for cargo delivery beyond the conventional secretory pathway. *C. neoformans* EVs have been described as a heterogeneous population of “virulence bags” containing numerous fungal survival and pathogenicity-associated cargo. Proteomic analyses of EV cargo have identified several cell surface glycoproteins, including members of the CDA family, well-known immunomodulators (Specht et al., 2017 [DOI](#); Rizzo et al., 2021 [DOI](#)). The ERQC system is essential for maintaining protein homeostasis by ensuring proper folding, glycosylation, and directed trafficking of secreted proteins to downstream compartments within the conventional secretory pathway. Defects in ERQC affect the fidelity of protein secretion not only by accumulation or degradation of misfolded proteins, leading to the impaired secretion of functional proteins, but also by the escape of misfolded forms, thus resulting in increased secretion of non-properly processed proteins (Marcus and Perlmutter, 2000 [DOI](#); Chen et al., 2024 [DOI](#)). However, the potential of ERQC to modulate EV-mediated protein trafficking is not yet systematically investigated.

This study presents the first systematic analysis of N-glycan-dependent ERQC in *C. neoformans* using mutant strains lacking ERQC gene homologs. Our findings highlight the critical roles of ERQC in maintaining cellular fitness and facilitating EV-mediated transport of virulence factors.

Results

Evolutionary unique features of *C. neoformans* ERQC components

We performed BLAST analysis of the *C. neoformans* H99 genome to identify the ERQC pathway-associated homologous genes of *C. neoformans*, followed by comparison with the ERQC components in other eukaryotic organisms (Supplementary Fig. S1, A). Unlike most eukaryotes, several yeast species within the Ascomycota phylum, such as *Saccharomyces cerevisiae* and *Candida albicans*, do not possess a functional UGGT. In contrast, UGGT homologs were identified in most Basidiomycota fungal species, including *C. neoformans* (Supplementary Fig. S1, A and B). The *C. neoformans* UGGT (CNAG_03648), which has been named as Ugg1, consists of 1,582 amino acids (aa) and features a signal peptide (1–20 aa) along with four tandem-like thioredoxin-like (TRXL) domains: TRXL12 (33–320 aa), TRXL13 (28–416 aa), TRXL14 (432–618 aa), and TRXL15 (712–950 aa). Ugg1 also contains a glucosyltransferase (GT) 24 domain with a DXD motif (1,369–1,371 aa), and a KDEL-like ER retention signal (1579–1582 aa), which facilitates its retrieval from the Golgi apparatus (Fig. 1B [DOI](#), Supplementary Fig. S2, A).

Additionally, we identified two *C. neoformans* ORFs, Mns1 (CNAG_02081) and Mns101 (CNAG_03240), as homologs of the eukaryote α 1,2-mannosidase I, which processes N-glycans before exporting them to the Golgi. The *C. neoformans* Mns1 and Mns101 show 42.4% and 28.8%

amino acid identities to *S. cerevisiae* ER α 1,2-mannosidase I, respectively, and share 30.9% identity between them. Furthermore, *C. neoformans* Mnl1 (CNAG_01987) and Mnl2 (CNAG_04498) were identified as putative components of ERQC in *C. neoformans* (Supplementary Fig. S1, A and C). *C. neoformans* Mnl1 is a homolog of the yeast α 1,2-mannosidase-like protein Htm1, which processes *N*-glycans targeted for ERAD. In contrast, *C. neoformans* Mnl2 encodes a mannosidase that does not share significant similarity with mannosidases from other eukaryotes. Mns1, Mns101, and Mnl1 possess glucosyl hydrolase (GH) 47 domains, essential for mannosidase activity, whereas Mnl2 contains a GH92 domain, also associated with mannosidase activity (**Fig. 1B**). The Mns1 and Mnl1 families are characterized by conserved cysteine (Cys) and alanine (Ala) residues within their activity domains, as previously described (Jakob et al., 2001). These conserved residues are present in *C. neoformans* Mns1, Mns101, and Mnl1 but are absent in Mnl2 (Supplementary Fig. S2, B). Notably, Mns101 and Mnl2 are Basidiomycota-specific proteins (Supplementary Fig. S1, A) and appear to have diverged early from other fungal mannosidase clades based on phylogenetic analysis (Supplementary Fig. S1, C). Interestingly, neither Mnl1 nor Mnl2 contains canonical KDEL/HDEL-like ER retention signals. In *S. cerevisiae*, the ER retention of Mnl1/Htm1 is mediated through its interaction with protein disulfide isomerase Pdi1, which carries an HDEL sequence (Gauss et al., 2011). Similarly, *C. neoformans* Mnl1 and Mnl2 may employ a non-canonical retention mechanism, likely facilitated by interactions with other ER-resident proteins, to achieve ER localization.

Loss of *UGG1*, *MNS1*, and *MNS101* causes alteration of *N*-glycan profiles in *C. neoformans*

We performed high-performance liquid chromatography (HPLC) analysis of the cell wall mannoproteins (cwMPs) from both the wild-type (WT) and ERQC mutant strains to investigate the ERQC malfunction-induced structural differences in *N*-glycans (**Fig. 2**). The HPLC profiles of cwMPs from the WT strain showed an M8 peak as the major species. This peak corresponded to *N*-glycans with 8 mannose residues (**Fig. 2A**, top). The glycan structure at the M8 peak (Man₈GlcNAc₂) in the WT primarily corresponded to the Man₇GlcNAc₂ core *N*-glycan with an additional mannose residue that is linked via an α 1,6-linkage and added in the Golgi apparatus (Park et al., 2012). In the *ugg1* Δ mutant, M8 was also the main *N*-glycan species, but the pools of hypermannosylated *N*-glycans (larger than the M11 peak) were markedly reduced (**Fig. 2A**, middle). The altered *N*-glycan profile of the *ugg1* Δ mutant was restored to that of the WT after complementation with the WT *UGG1* gene (**Fig. 2A**, bottom). The lectin blotting analysis using *Galanthus nivalis* agglutinin (GNA), which specifically binds to terminal α 1,2-, α 1,3-, and α 1,6-linked mannose residues, showed a distinctive increase of glycoproteins with lower molecular weight in the secretory proteins in *ugg1* Δ compared to those in WT (**Fig. 2B**). This observation aligns with the decrease of hypermannosylated *N*-glycans in the *ugg1* Δ mutant. Overall, these results suggest that Ugg1 is involved in mediating the hypermannosylation of *N*-glycans in the Golgi apparatus.

We also assessed putative α 1,2-mannosidase I genes for roles in protein mannosylation. The *mns1* Δ mutant *N*-glycan profile showed a peak shift from an M8 to an M9 form, which strongly indicates that *C. neoformans* Mns1 could be the primary ER α 1,2-mannosidase I (**Fig. 2C**, *mns1* Δ). Notably, the loss of Mns101, which is present only in Basidiomycota, increased the fractions containing hypermannosylated glycans (> M10) while maintaining M8 as a primary core *N*-glycan form. This suggests that the basidiomycete-specific Mns101 may potentially be a novel α 1,2-mannosidase that functions to remove mannose residues from hypermannosylated *N*-glycans in the Golgi apparatus or further trims the M8 glycan in the ER before the glycoproteins are transported to the Golgi (**Fig. 2C**, *mns101* Δ). The matrix-assisted laser desorption ionization-time of flight (MALDI-TOF) mass spectrometer analysis of neutral *N*-glycans (**Fig. 2C**), further confirmed the shift of the major M8 peak in the WT strain to M9 in the *mns1* Δ mutant and the increase in the hypermannosylation profile in the *mns101* Δ mutant (**Fig. 2D**). The *N*-glycan profiles of the *mns1* Δ *mns101* Δ double mutant showed the combined effect of each null mutation, and it showed both increased

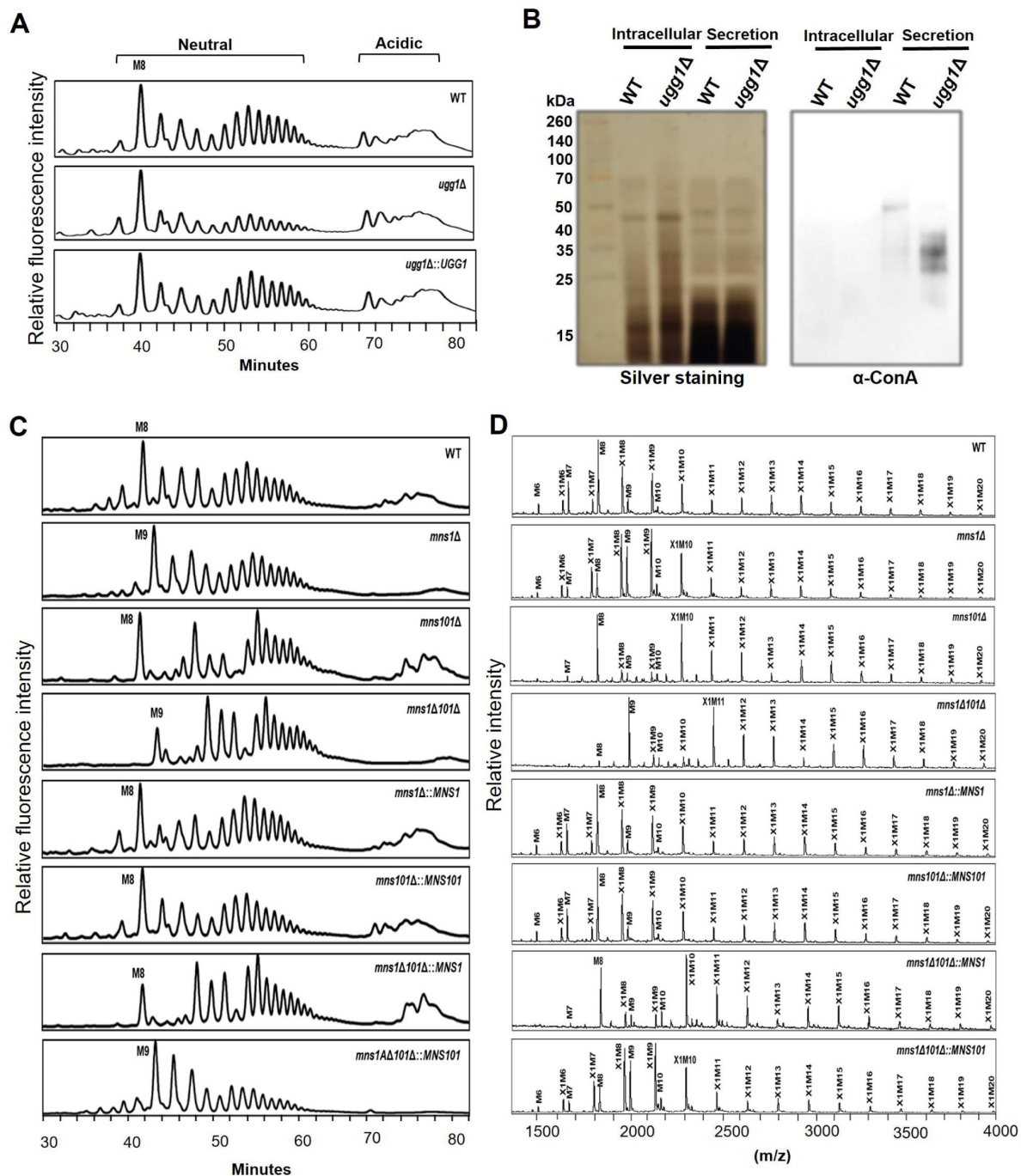


Figure 2.

N-glycan profile analysis of *C. neoformans* ERQC mutant strains.

The HPLC and MALDI-TOF-based *N*-glycan structure analysis were carried out as described in Supplementary information. (A) HPLC-based analysis of *N*-glycan profiles of the *ugg1Δ* mutant. (B) Lectin blotting of sodium dodecyl sulphate (SDS)-polyacrylamide gels containing intracellular or secreted proteins into the culture supernatants of the wild type (WT) and *ugg1Δ* strains. Yeast cells were cultivated in YPD medium for 24 h, harvested, and subjected to sample preparation of soluble intracellular proteins and secreted proteins. The proteins (30 μg) were loaded on 15% SDS-polyacrylamide gel and analyzed using silver staining (left) or blotting (right) with *Galanthus nivalis* agglutinin conjugated to horseradish peroxidase (GNA-HRP, Roche). (C) HPLC analysis of total *N*-glycan profiles of *mns1Δ*, *mns101Δ*, and *mns1Δ101Δ* mutants. (D) MALDI-TOF profiles of neutral *N*-glycans of *mns1Δ*, *mns101Δ*, and *mns1Δ101Δ* mutants. The *N*-glycans of cell wall mannoproteins from *C. neoformans* cells were AA-labelled and analyzed using HPLC. For MALDI-TOF analysis, neutral *N*-glycan fractions were obtained from the HPLC fractionation of total *N*-glycans.

hypermannosylated glycans and a shift from the M8 to the M9 peak in both the HPLC (**Fig. 2C** [↗](#), *mns1Δ101Δ*) and MALDI-TOF analyses. This indicates that Mns1 and Mns101 serve as mannosidases and play independent roles at different stages of *N*-glycan processing in *C. neoformans*.

In contrast, the loss of both *MNL1* and *MNL2* did not show notable differences in the HPLC profile of *N*-glycans from cell surface mannoproteins (Supplementary Fig. S3, A). Considering the expected function of Mnl1 and Mnl2 in ERAD, their substrates are likely misfolded proteins that have not been transported to the Golgi apparatus. Thus, we speculate that only the normally processed *N*-glycan profiles of cell surface mannoproteins in the *mnl1Δ mnl2Δ* mutant strain were observed.

Loss of ERQC components results in defective growth fitness and increased stress sensitivity

To investigate the changes in ERQC-related gene expression under stress conditions, we conducted quantitative reverse transcription PCR (qRT-PCR) analysis of *C. neoformans* following treatment with tunicamycin (TM, 5 μg/ml), a well-established inhibitor of *N*-glycosylation, or dithiothreitol (DTT, 20 mM), which disrupts disulfide bond formation and induces the accumulation of misfolded proteins. Additionally, we examined gene expression after growth at 37 °C (**Fig. 3A** [↗](#)). When misfolded proteins accumulate in the ER, the unfolded protein response (UPR) system is activated to restore homeostasis (Chakraborty et al., 2016 [↗](#)). As expected, *KAR2*, encoding a molecular chaperone associated with the UPR system, was upregulated in response to TM treatment, DTT treatment, and heat stress at 37 °C. The ERQC components *UGG1*, *MNS1*, *MNS101*, *MNL1*, and *MNL2* were upregulated by DTT treatment; however, only *UGG1* was slightly induced following TM treatment. The high-temperature conditions also induced the expression of *KAR2*, *UGG1* and *MNL2*. Notably, DTT treatment induced a 10-fold higher expression of *MNS1* and *MNL1*. These results strongly suggest that the *C. neoformans* genes *UGG1*, *MNS1*, *MNS101*, *MNL1*, and *MNL2* are crucial components of the ERQC and ERAD systems, although each respond uniquely to various cell/ER stress conditions.

We next investigated the growth of the *ugg1Δ* mutant under various stress-inducing conditions to elucidate the roles of the ERQC components in ER stress response and adaptation, which are closely associated with *C. neoformans* virulence. The *ugg1Δ* mutant showed impaired growth even under normal growth conditions (**Fig. 3B** [↗](#)) and increased sensitivity to ER stress-inducing agents such as DTT, TM and cell wall stressors, including as calcofluor white (CFW), congo red (CR), sodium dodecyl sulfate (SDS), and caffeine. Additionally, *ugg1Δ* showed hindered growth in the presence of antifungal drugs such as azole agents (fluconazole and ketoconazole) and the glucose transport inhibitor fludioxonil. These findings suggest that Ugg1 is crucial for the robust growth and survival of *C. neoformans* under various stress conditions. In contrast, the single deletion (*mns1Δ* and *mns101Δ*) and double deletion (*mns1Δ101Δ*) strains did not exhibit noticeable phenotypic changes except for a slight increase in sensitivity of *mns101Δ* and *mns1Δ101Δ* to higher fludioxonil concentrations (Supplementary Fig. S3, B). Similarly, neither the single nor double disruption of *MNL1* and *MNL2* produced detectable changes in the tested growth conditions (Supplementary Fig. S3, C).

Amino acid analogues have been used for evaluating the functionality of protein folding, as its incorporation into newly synthesized proteins interrupts proper protein folding (Trotter et al., 2002 [↗](#); Cowie et al., 1959 [↗](#)). We hypothesized that, in the context of a defective ERQC pathway, misfolded proteins cannot be adequately repaired and accumulate, thus triggering ER stress, which may ultimately inhibit cell growth in the presence of amino acid analogues. In the presence of the leucine analogue 5',5',5'-trifluoroleucine (TFL), *ugg1Δ* showed noticeably inhibited growth, whereas the WT strain showed no growth inhibition (**Fig. 3C** [↗](#)). The *mns1Δ101Δ* strain also exhibited slightly increased sensitivity to TFL compared with that of the WT strain.

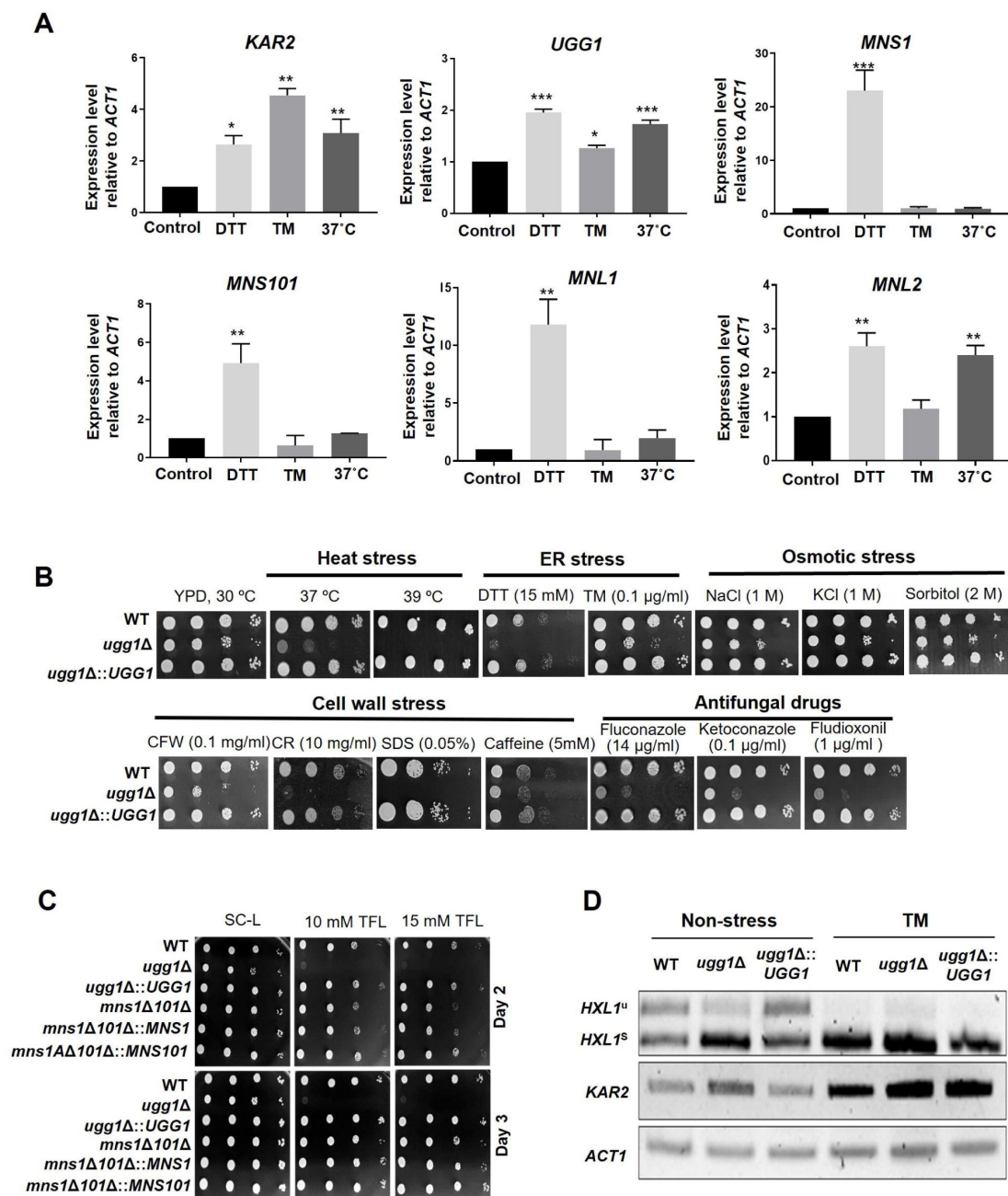


Figure 3.

Growth phenotype of *C. neoformans* ERQC mutant strains.

(A) Expression analysis of ERQC genes in *C. neoformans*. Yeast cells were cultured in YPD medium to a mid-logarithmic phase and exposed to dithiothreitol (DTT; 20 mM), tunicamycin (TM; 5 μ g/ml) or cultured at 37 $^{\circ}$ C for 1 h. The relative transcript levels of *C. neoformans* genes were analyzed using qRT-PCR and normalized with that of *ACT1*. Error bars represent standard deviation of duplicated assays. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test. *** $P < 0.0005$, ** $P < 0.003$, * $P < 0.05$. (B) Spotting analysis of *C. neoformans* *ugg1* Δ mutant strains under various stress conditions such as heat stress (37 $^{\circ}$ C and 39 $^{\circ}$ C), ER stress (DTT and TM), cell-wall stress (CFW: calcofluor white, CR: Congo red, SDS: Sodium dodecyl sulfate, caffeine), osmotic stress (NaCl, KCl, sorbitol) and treatment with antifungal drugs (fluconazole, ketoconazole, fludioxonil). (C) Growth analysis in the presence of 5',5',5'-trifluoroleucine (TFL). Respective strains were spotted on SC-Leucine media with or without TFL supplementation. Plates were incubated for 3 days at 30 $^{\circ}$ C. (D) RT-PCR analysis of *IRE1*-dependent splicing of *HXL1*. Strains were cultured in YPD supplemented with 5 μ g/ml TM.

As a defense mechanism against misfolded protein accumulation-mediated ER stress, UPR is induced by the unconventional splicing of the *HXL1* transcription factor in *C. neoformans* (Cheon et al., 2011 [DOI](#)). RT-PCR analysis revealed a significantly elevated level of spliced *HXL1* in the *ugg1Δ* strain compared to the WT, even under normal culture conditions (**Fig. 3D** [DOI](#)), strongly indicating that the loss of *UGG1* leads to the accumulation of misfolded proteins, triggering ER stress under standard growth conditions. Furthermore, green fluorescence protein (GFP)-tagged Ugg1, Mns1, and Mns101 proteins colocalized with the ER marker, supporting their role as confirming them to be functional ERQC components based on their subcellular localization in the ER (Supplementary Fig. S4).

ERQC defects lead to virulence attenuation in *C. neoformans*

We investigated the effects of a defective ERQC on *in vitro* virulence phenotypes of *C. neoformans* by analyzing capsule and melanin production, as these are two of the major virulence factors of *C. neoformans*. Culturing on L-DOPA-containing plates showed that melanin production was reduced in the *ugg1Δ* strain compared with that in the WT (**Fig. 4A** [DOI](#), top), whereas the *mns1Δ101Δ* mutant cells did not show detectable defects at either 30 or 37 °C (**Fig. 4A** [DOI](#), bottom). Considering the decreased growth of the *ugg1Δ* strain, the melanin production of *C. neoformans* cells cultivated in liquid L-DOPA was measured and normalized by cell density, which further confirmed the decreased melanin production activity of the *ugg1Δ* strain (**Fig. 4B** [DOI](#)). India ink staining of the capsule showed significantly reduced capsule thickness in the *ugg1Δ* strain and a moderate defect in the *mns1Δ101Δ* strain (**Fig. 4C** [DOI](#)).

Defects in the ERQC system led to a more apparent decrease in pathogenicity in a murine model of systemic cryptococcosis. Notably, the *ugg1Δ* strain was almost avirulent with no detectable signs of illness in the infected animals (**Fig. 4D** [DOI](#)). Although *MNS1* and *MNS101* single disruptions did not cause a detectable decrease in *C. neoformans* pathogenicity, their double disruption resulted in decreased virulence. Histopathological analysis of infected mice lungs suggested significantly reduced lung colonization by both *ugg1Δ* and *mns1Δ101Δ* cells (Supplementary Fig. S5, A). Additionally, the fungal burden of *ugg1Δ*-infected animals showed a notable decrease in organ colonization at 60 days post-infection (dpi) (Supplementary Fig. S5, B). Analysis of the fungal burden at 7 dpi showed significantly reduced organ colonization of the *mns1Δ101Δ* mutant compared with that of the WT (Supplementary Fig. S5, C). Consistent with the *in vivo* survival data, the number of surviving *ugg1Δ* and *mns1Δ101Δ* cells within the macrophage-like cell line J447A.1 was significantly lower compared with that of the WT strain (**Fig. 4E** [DOI](#)). These data suggest that a functional ERQC system is critical for maintaining full *in vivo* pathogenicity and ensuring robust survival within host immune cells.

Loss of Ugg1 function manifests defects in extracellular transport of GXM

Cryptococcal capsule materials are synthesized intracellularly and secreted to the cell surface. Subsequently, they are assembled and bound to the cell wall (Yoneda and Doering, 2006 [DOI](#)). The observed capsule defects in the *ugg1Δ* and *mns1Δ101Δ* mutant strains may be attributed to issues in the capsule polysaccharide synthesis or transport steps. Alternatively, defective capsule formation could arise from an increased shedding of capsule polymers, preventing their proper attachment to the cell surface. To determine the cause of the capsule defects, we examined the amounts of capsular polysaccharides synthesized intracellularly and those shed into the culture supernatant by a capsule blotting assay, using the 18B7 antibody directed against GXM, the major capsule component (Casadevall et al., 1998 [DOI](#)). As controls, we also tested capsule production and secretion in the *cap59Δ* strain with a defect in GXM synthesis (Grijpstra et al., 2009 [DOI](#)) and the *rim101Δ* strain with a capsule attachment defect despite normal polysaccharide synthesis and secretion (O'Meara et al., 2010 [DOI](#)). Intracellular GXM quantities did not notably differ between the WT, *ugg1Δ*, *mns1Δ101Δ*, and *rim101Δ* strains, although *ugg1Δ* showed slightly reduced GXM level (**Fig. 5A** [DOI](#), left). In contrast, no intracellular GXM was detected in *cap59Δ*, which was consistent

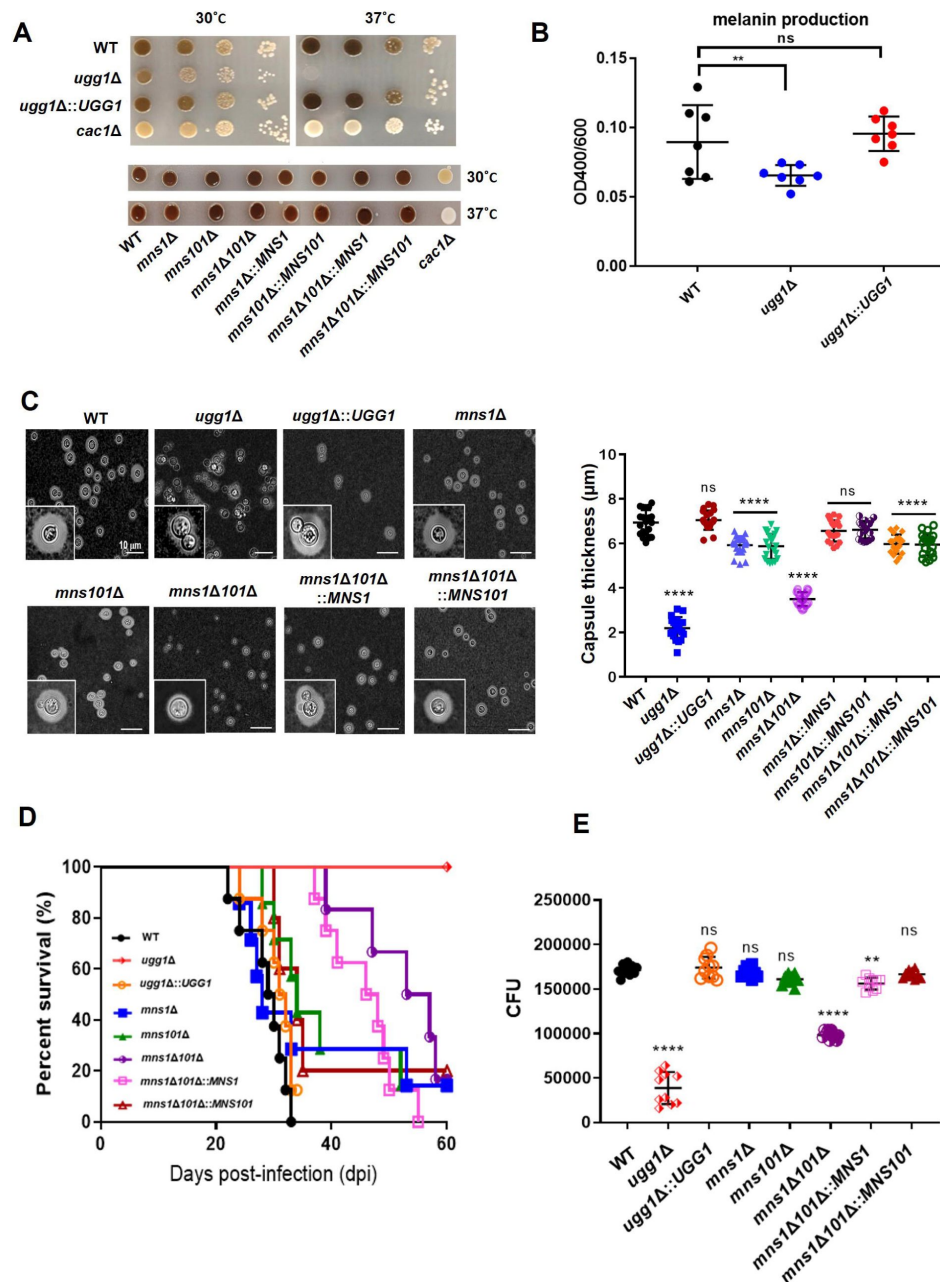


Figure 4.

***In vitro* and *in vivo* virulence-associated phenotypes of *C. neoformans* UGG1, MNS1, and MNS101 mutant strains.**

(A) Melanin synthesis analysis on L-DOPA plates. WT, $ugg1\Delta$, $ugg1\Delta::UGG1$, $mns1\Delta$, $mns101\Delta$, $mns1\Delta101\Delta$, $mns1\Delta101\Delta::MNS1$, $mns1\Delta101\Delta::MNS101$, and $cac1\Delta$ (negative control) strains were serially diluted, plated on L-DOPA plates, and incubated at 30 °C and 37 °C. (B) Melanin synthesis activity per cell of the $ugg1\Delta$ mutant. WT, $ugg1\Delta$, $ugg1\Delta::UGG1$ were cultured in liquid L-DOPA medium. The amount of melanin in the culture supernatant was measured and normalized by cell density. Capsule formation. Cells were cultured for 2 days in 10% Sabouraud media at 30 °C and observed under the microscope. Statistical significance: ****, $P < 0.0001$, ns, not significant). *In vivo* virulence analysis. A/Jcr mice ($n=8$) were infected with 10^5 cells of WT, $ugg1\Delta$, and $ugg1\Delta::UGG1$, $mns1\Delta$, $mns101\Delta$, $mns1\Delta101\Delta$, $mns1\Delta101\Delta::MNS1$, and $mns1\Delta101\Delta::MNS101$ strains, and survival was monitored for 2 months, as described in the Supplementary Information. (E) Survival of *C. neoformans* in macrophages. Survival of *C. neoformans* cells within the J774A.1 macrophage-like cell line was determined by counting colony formation unit (CFU) obtained from lysed macrophages from two biologically independent experiment sets. **** $P < 0.0001$, *** $P < 0.0005$, * $P < 0.05$, ns, not significant. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test.

with its defect in GXM synthesis. Moreover, the *ugg1Δ* strain was defective in capsule polymer secretion/shedding, whereas the *mns1Δ101Δ* strain displayed enhanced shedding of GXM-containing polymers that appeared to be shorter in length than the WT-type polymers, suggestive of poor polysaccharide polymerization (Fig. 5A, right). Therefore, these results suggest that *ugg1Δ* contains a defect in polysaccharide trafficking to the extracellular space, whereas *mns1Δ101Δ* has a defect in polysaccharide polymerization, resulting in defective capsule elaboration, albeit with differential degrees of severity in both ERQC mutants.

Furthermore, to determine whether the ERQC mutants are also impaired in capsule attachment, we performed a capsule transfer assay, to assess whether exogenously shed capsule polysaccharides can bind to acapsular mutants (Reese and Doering, 2003). The hypocapsular *rim101Δ* mutant did not show reassociation of capsular polysaccharides; however, the acapsular *cap59Δ* mutant showed recovery of capsule formation when co-incubated with exogenous GXM (Fig. 5B). Similarly, *ugg1Δ* reverted its acapsular phenotype on incubation with the WT-shed polysaccharides, strongly suggesting that the defective capsule phenotype of *ugg1Δ* is primarily due to defective GXM secretion rather than impaired attachment of capsule polymers. We further conducted a capsule transfer assay using capsular polysaccharides shed from *mns1Δ101Δ*. The *cap59Δ* mutant reverted to the capsular phenotype by attaching to the *mns1Δ101Δ*-shed polymers. However, the fluorescence intensity of the capsule generated was significantly lower than that of the cells attached with polysaccharides obtained from the WT. Although the decreased length of capsule polysaccharides should be validated by techniques specifically measuring GXM size (De Jesus et al., 2010), this result suggests that the *mns1Δ101Δ* double mutant secretes incomplete capsule polysaccharides (Fig. 5C), leading to a hypocapsular phenotype. These results collectively indicate that disrupting the ERQC results in impaired capsule formation by defective GXM trafficking to the extracellular space.

Transmission electron microscopy (TEM) imaging of the WT and mutant cells distinctly showed a diminished capsule structure and loss of capsule shedding in the *ugg1Δ* strain compared with that of the WT strain (Fig. 5D). Additionally, we observed considerable thinning of the cell walls in both ERQC mutant cells. Notable changes in intracellular structures along with an increase in the number of pigmented vesicles were observed in the mutant cells even under YPD culture conditions. Under capsule-inducing conditions, a significant accumulation of vesicular structures (electron-lucent structures) was observed intracellularly, particularly in *ugg1Δ*. We performed lipid droplet (LD) staining to determine whether these vesicular structures might be LDs. We observed a notable increase in LDs in *ugg1Δ* under capsule-inducing culture conditions (Supplementary Fig. S6, A). LDs impact proteostasis by sequestering misfolded proteins intended for degradation and providing an “escape hatch” when the ERQC is overloaded (Ploegh, 2007; Vevea et al., 2015). Additionally, FM4-64 dye staining showed increased number of vacuoles in the *ugg1Δ* cells compared with that in the WT (Supplementary Fig. S6, B). Moreover, all the observed abnormal phenotypes were less pronounced in the *mns1Δ101Δ* mutant than in the *ugg1Δ* strain. Collectively, the altered vesicular structures observed in the *ugg1Δ* and *mns1Δ101Δ* strains highlight the critical role of ERQC in maintaining proper vesicular architecture and cell surface organization. These functions are tightly linked to extracellular trafficking of virulence factors and cell wall remodeling.

***ugg1Δ* transcriptomic profiling shows induced ER and cell wall integrity stress responses**

To elucidate the mechanisms underlying *UGG1* deletion-induced physiological changes in *C. neoformans*, we performed RNA sequencing (RNAseq)-based transcriptome analysis of the WT and *ugg1Δ* strains under standard growth conditions (YPD medium, 30 °C). Comparative transcriptome analysis showed statistically significant alterations exceeding 2-fold in the expression patterns of 146 genes, of which 85 were upregulated and 61 were downregulated compared with that of the WT strain (Fig. 6A, B, Supplementary Tables S2, A and B).

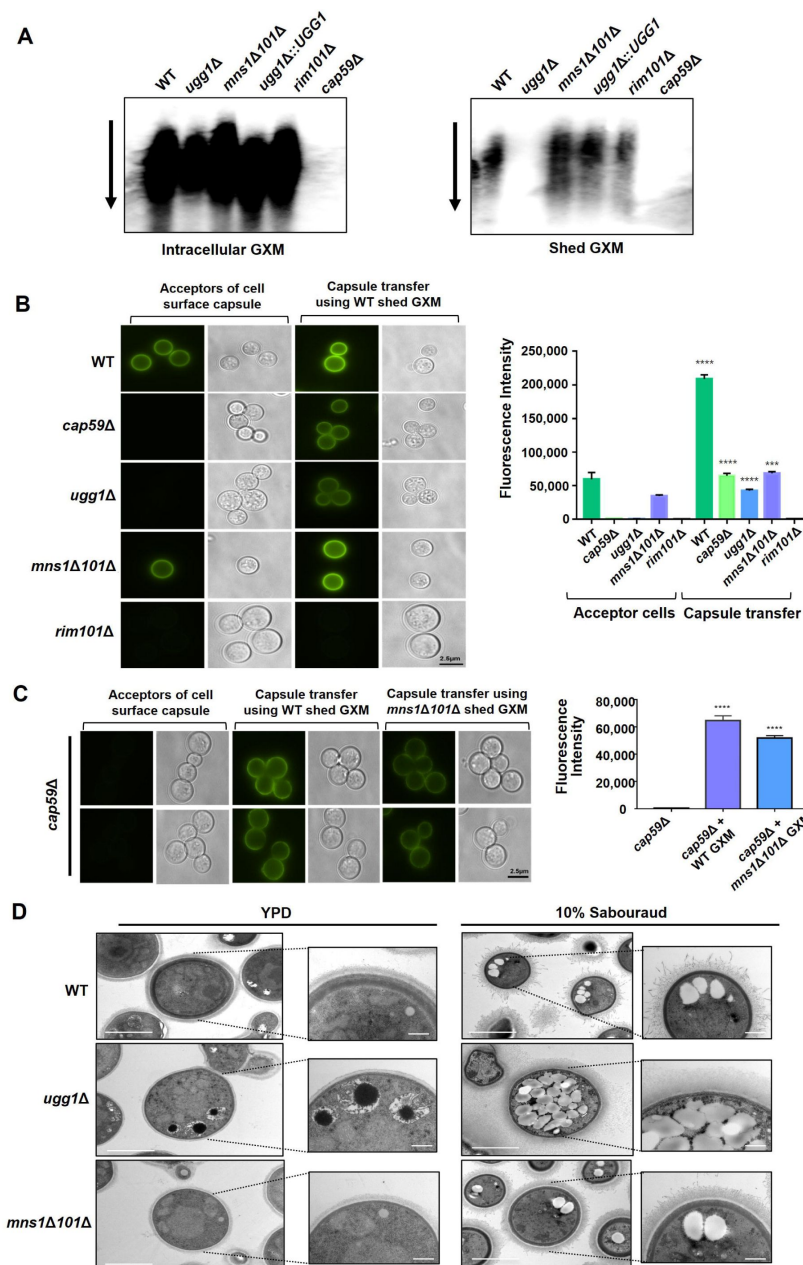


Figure 5.

Capsule shedding and transfer analysis of *C. neoformans* *UGG1*, *MNS1*, and *MNS101* mutant strains.

(A) Capsule shedding analysis. The presence of intracellular (left) and shed (right) glucuronoxylomannan (GXM) was assessed by blotting a cell culture filtrate using the monoclonal antibody 18B7. The arrow indicates the direction of electrophoresis. (B) Capsule transfer analysis using exogenous capsule material from WT. The capsule transfer assay was performed using the indicated strains as acceptors. Surface capsules were probed using the anti-GXM antibody 18B7 conjugated with AlexaFluor 488. Quantitative measurement of fluorescence intensity was calculated based on independent triplicate experiments with standard deviations presented as error bars. (C) Capsule transfer analysis using exogenous capsule material from WT or *mns1Δ101Δ*. Statistical significance: **** $P < 0.0001$. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test. (D) Transmission electron microscopy (TEM) of *C. neoformans* WT, *ugg1Δ*, and *mns1Δ101Δ* strains. Yeast cells were grown overnight at 30 °C in YPD medium and fixed in 2% glutaraldehyde and 2% paraformaldehyde. A Zeiss Axioscope (A1) equipped with an AxioCan MRm digital camera was used to visualize India ink-stained *C. neoformans* cells. Specimens were prepared using critical point drying prior to TEM microscopy. Capsule and yeast cell body diameters were measured using ImageJ (National Institute of Health).

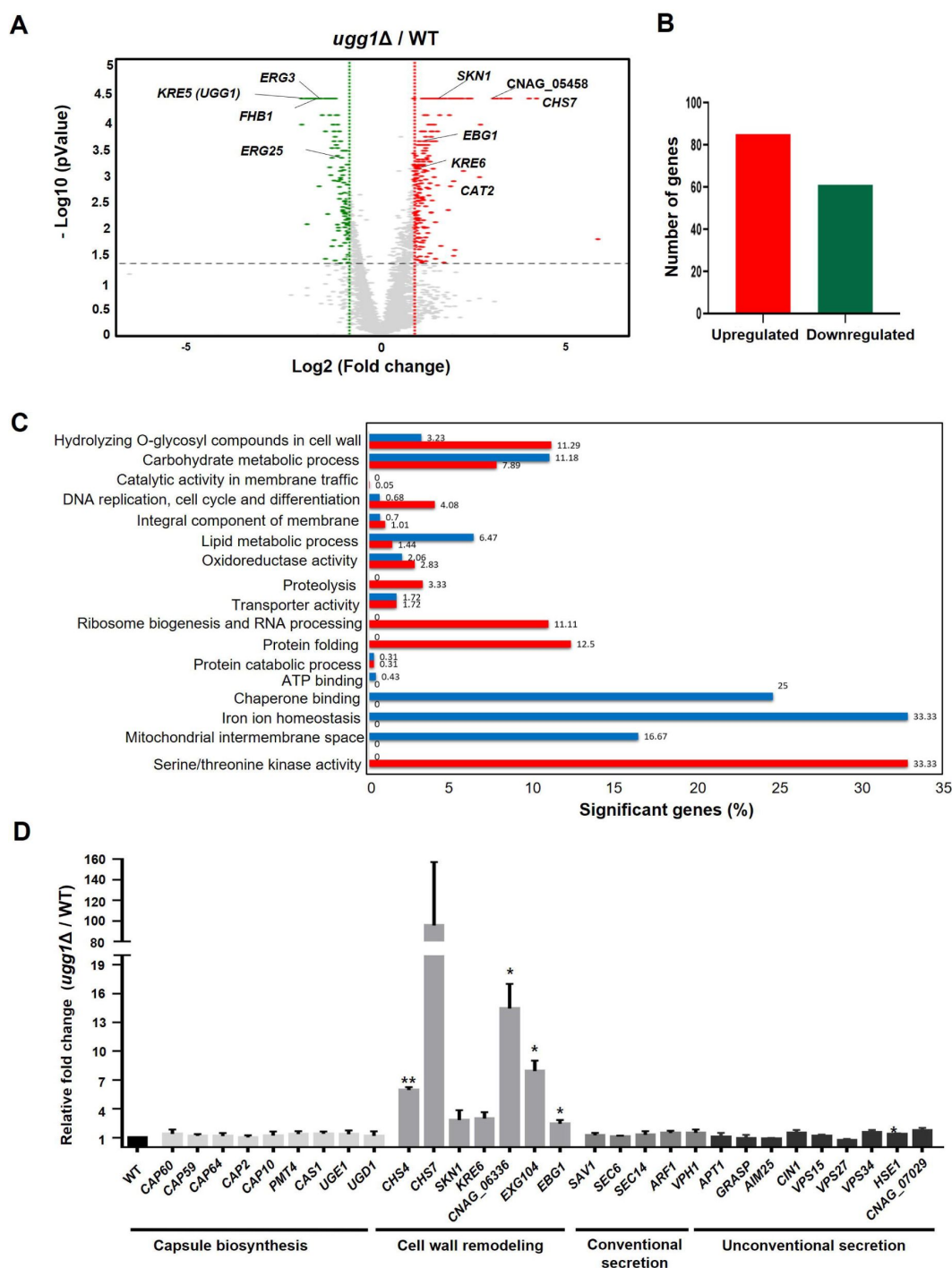


Figure 6.

Transcriptome analysis of *C. neoformans* WT and *ugg1Δ* cells.

(A) Volcano plot comparing a 2-fold differential gene expression between *ugg1Δ* and WT strains under normal growth conditions. (B) Number of genes upregulated and downregulated by ≥ 2 -fold in *ugg1Δ* compared with that of the WT. (C) Gene ontology (GO) analysis of differentially expressed genes between WT and *ugg1Δ* strains. Significantly upregulated genes in *ugg1Δ* are shown in red, whereas significantly downregulated genes in *ugg1Δ* are shown in blue. Total RNA was extracted and subjected to RNAseq analysis as described in the Supplementary Information. (D) qRT-PCR analysis of mRNA expression levels of a set of genes responsible for capsule biosynthesis, cell wall remodeling, and both conventional and non-conventional secretion in *ugg1Δ* vs WT under normal growth conditions from three biologically independent experiment sets.

Notably, the *ugg1Δ* mutant showed upregulation of genes such as *SKN1* and *KRE6* (transmembrane glucosidases involved in the sphingolipid biosynthesis and β -glucan biosynthesis), *CAT2* (a putative peroxisomal catalase), *CHS7* (chitin synthase export chaperone), and *CNAG_05458* (a putative endo-1,3(4)- β -glucanase). In contrast, genes encoding *ERG3* and *ERG25* (associated with the ergosterol biosynthesis pathway) and *FHB1* (flavo-hemoglobin denitrosylase associated with counteracting nitric oxide stress) were significantly downregulated. Gene Ontology (GO) analysis in *ugg1Δ* showed pronounced induction of genes implicated in various cellular processes, including the hydrolysis of *O*-glycosyl compounds for cell wall remodeling, DNA replication, proteolysis, ribosome biogenesis, protein folding, and serine/threonine kinase activity. In contrast, genes associated with carbohydrate and lipid metabolism, chaperone binding, iron homeostasis, and mitochondrial intermembrane space were considerably downregulated (**Fig. 6C**). The transcriptome profile strongly suggests that loss of Ugg1 function induces the expression of several genes involved in maintaining cell wall integrity to compensate for cell wall defects, particularly those associated with chitin (*CDA2*, *CHS7*, *QRI1*, and *CNAG_06898*) and glucan biosynthesis (*SKN1*, *KRE6*, *EBG1*, *LPI9*, *CNAG_05458*, and *BLG2*). Furthermore, genes coding for ER chaperones that aid in protein folding (*ERO1*, *KAR2*, *LHS1*, and *PDI1*) and chaperone regulator (*SCJ1*), and genes involved in proteolysis in the ER (*CNAG_04635*, *CNAG_06658*) were induced, likely as part of the ER stress response induced by the accumulation of misfolded proteins in the presence of a defective ERQC.

Further investigation of the effects of *UGG1* disruption in the expression of genes involved in capsule biosynthesis, cell wall remodeling, and the secretion pathways of protein virulence factors using qRT-PCR analysis (**Fig. 6D**) showed no noticeable changes in the expression of genes related to capsule biosynthesis and conventional or unconventional secretion pathways at mRNA level, despite the aberrantly displayed defective phenotypes.

EV-mediated protein trafficking and non-conventional secretion is impaired in the *ugg1Δ* mutant strain

Many enzymes contribute to the composite cryptococcal virulence phenotype. Some of these virulence-associated enzymes are secreted through traditional secretion pathways, whereas others are packaged into EVs and released into the extracellular milieu via non-conventional secretion mechanisms (Almeida et al., 2015). The defects observed in the ERQC mutant strains, particularly in the production of melanin and polysaccharide capsules, both of which are transported by EVs, suggest a possible impairment in the EV-mediated extracellular transport of virulence-associated enzymes in these mutants. Therefore, we examined the *in vitro* activity of urease, and laccase and acid phosphatase in both intracellular and extracellular fractions.

Comparative analysis of the WT and *ugg1Δ* strains showed minimal changes in the intracellular activities of urease, laccase, and acid phosphatase. However, their activities were significantly reduced in the extracellular fractions of the *ugg1Δ* cells (**Fig. 7A**, **7B**, **7C**, and **7D**). Importantly, urease activity was performed in both solid and liquid media, demonstrating a notable loss of secreted urease activity in *ugg1Δ* cells, with a slight decrease in the *mns1Δ101Δ* double mutant cells (**Fig. 7A** and **7B**). As urease lacks a signal peptide required for conventional protein secretion, reduced urease secretion strongly indicates a compromised EV-mediated non-conventional secretory pathway in the *ugg1Δ* mutant strain. The decrease in extracellular laccase activity (**Fig. 7C**) corroborates our melanization assay results, in which *ugg1Δ* showed defective melanin production (**Fig. 4A**).

We next examined cellulase and α -amylase activities to evaluate the impact of ERQC disruption on the conventional secretion pathway of non-virulence-related enzymes (**Fig. 7E** and **7F**). The extracellular activities of these carbohydrate polymer-degrading enzymes in the *ugg1Δ* mutant were slightly lower than those observed in the WT, likely reflecting reduced intracellular enzyme production rather than a substantial impairment of secretion efficiency. Additionally, we analyzed

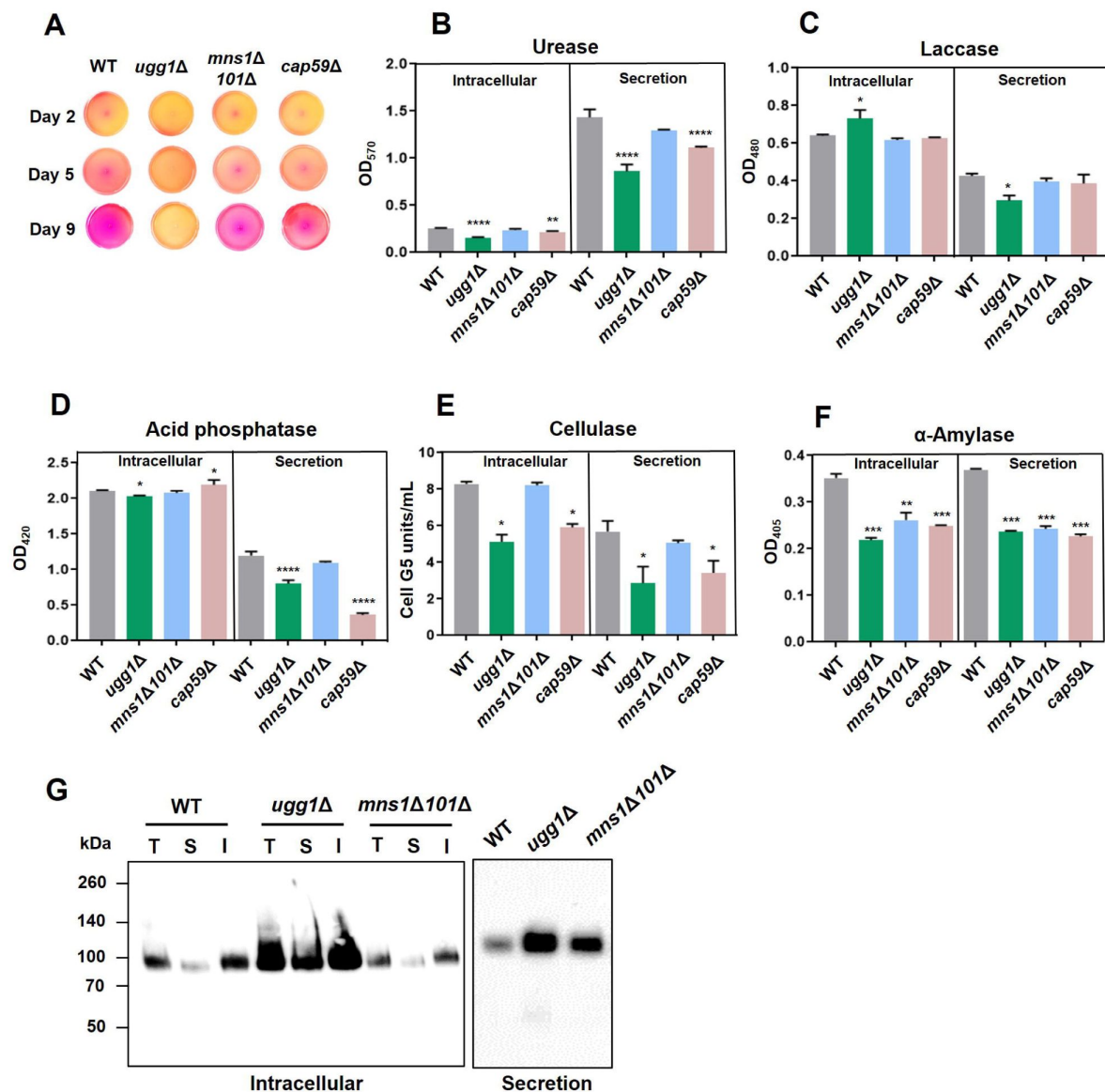


Figure 7.

Analysis of protein secretion in *C. neoformans* UGG1, MNS1, and MNS101 mutant strains.

(A) Spot assay for urease analysis on Christensen's urea agar. Absence of pink coloration indicates loss of urease activity. (B, C, D) Analysis of secretion for virulence-related enzymes such as urease, laccase, and acid phosphatase from three biologically independent experiment sets. Statistical significance: **** $P < 0.0001$, ** $P < 0.003$, *** $P < 0.005$, * $P < 0.05$. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test. (E, F) Analysis of secretion for non-virulence-related enzymes such as cellulase and α -amylase. Statistical significance: *** $P < 0.0005$, ** $P < 0.003$, * $P < 0.05$, ns, not significant. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test. (G) Analysis of the conventional secretion of Cda1 in *C. neoformans*. Presence of Cda1 was analyzed in total (T), soluble (S), and insoluble (I) fractions of intracellular extracts (left), along with the secreted fraction (right). Subcellular fractionations were performed as previously described (Thak et al., 2022), and the fractions were subjected to western blotting analysis using an anti-Cda1 antibody.

the localization of chitin deacetylase I (Cda1), a glycosylphosphatidylinositol (GPI)-anchored protein involved in converting chitin to chitosan, essential for maintaining cell wall integrity (Baker et al., 2007 [↗](#); Baker et al., 2011 [↗](#); Upadhyaya et al., 2018 [↗](#); Upadhyaya et al., 2021 [↗](#)). Cda1 was primarily detected in the insoluble cellular protein fraction, encompassing the cell wall and membrane compartments (Fig. 7G [↗](#), left), consistent with its cell surface localization via a GPI anchor. Cda1 is secreted extracellularly upon cleavage of the GPI anchor. Notably, the *ugg1Δ* mutant exhibited significantly higher intracellular and extracellular Cda1 levels compared to the WT and *mns1Δ101Δ* strains (Fig. 7G [↗](#), right), aligning with a 1.87-fold increase in *CDA1* mRNA expression observed in the RNA-seq data. These findings indicate that the surface localization and secretion of Cda1 remain efficient, suggesting that the conventional secretion pathway is largely unperturbed in *ugg1Δ*. Collectively, these results suggest that while Ugg1-mediated ERQC defects have a pronounced negative effect on EV-mediated protein transport, their impact on the conventional secretion pathway is minimal.

Extracellular vesicle biogenesis and cargo loading are defective in *ugg1Δ* strain

We thus analyzed the number and size distribution of EVs to determine possible abnormal EV-mediated trafficking of virulence factors in the *ugg1Δ* mutant. Nanoparticle tracking analysis (NTA) showed a major peak in size distribution at approximately 150 (134±28) nm in the WT strain, which was similar in size to mammalian exosomes and typical microbial EVs. Additionally, a minor peak ranging from 300–500 nm was observed and corresponded to microvesicles (Fig. 8A [↗](#)). Notably, the size distribution of *ugg1Δ* EVs was more heterogeneous with smaller EVs ranging from 50–150 nm (82±16; 124±14) compared with that of the WT. The *cap59Δ* EVs displayed a major distribution of approximately 150 nm (120±21). Microvesicles were barely detected in either *cap59Δ* or *ugg1Δ* strains.

We observed a significant reduction in the total number of secreted EVs in the *ugg1Δ* mutant (approximately 40%), suggesting defective EV biogenesis and/or stability in the absence of functional Ugg1 (Fig. 8B [↗](#)). Cryo-TEM analysis of EV morphology confirmed that the sizes of EVs released by *ugg1Δ* were significantly smaller and more diverse than those of the WT or the acapsular mutant (Fig. 8C [↗](#)), further supporting the NTA results. Examining EV size distribution by measuring their diameter (Fig. 8D [↗](#)) further confirmed that although the EVs in the WT strain ranged from approximately 50–550 nm in size with distribution primarily concentrated at approximately 150 nm, the EVs in the *ugg1Δ* mutant were smaller (< 100 nm) with heterogeneous sizes. Notably, *cap59Δ* EVs were uniformly distributed at approximately 150 nm and were present at a higher concentration than those of EVs from the WT. An increase in EV release and virulence factors has been also observed in acapsular mutant strains of *C. neoformans* and *C. gattii* (Rodrigues et al., 2007 [↗](#); Reis et al., 2019 [↗](#)), supporting the notion that the capsule serves as a barrier in EV release in capsule forming species.

To characterize the protein composition of EVs from WT and *ugg1Δ* strains, EV-associated and whole-cell lysate (WCL) proteins were extracted and subjected to proteomic analysis (Fig. 8E [↗](#)). In the WCL, 4,678 proteins were identified in the WT strain, with 333 exhibiting differential expression (>2-fold) in *ugg1Δ* versus the WT strain, including 123 upregulated and 210 downregulated proteins (Supplementary Fig. S7, A, left). In EVs, 2,075 proteins were detected in the WT strain, of which 693 were differentially expressed in EVs from *ugg1Δ* relative to EVs from the WT strain (>2-fold), comprising 273 upregulated and 420 downregulated proteins (Supplementary Fig. S7, A, right; Supplementary Table S3). Additionally, comparative proteomic analysis revealed 180 proteins uniquely enriched in EVs (Supplementary Fig. S7, B). The GO enrichment analysis of differentially expressed proteins (> 2-fold) showed distinctive patterns in biological process (BP) and cellular components (CC) between EVs and the WCL (Supplementary Fig. S7, C). Three proteomic analyses of *C. neoformans* EVs have been reported previously (Rodrigues et al., 2008 [↗](#); Wolf et al., 2014 [↗](#); Rizzo et al., 2021 [↗](#)), identifying 76, 202, and 1,847 proteins associated with EVs,

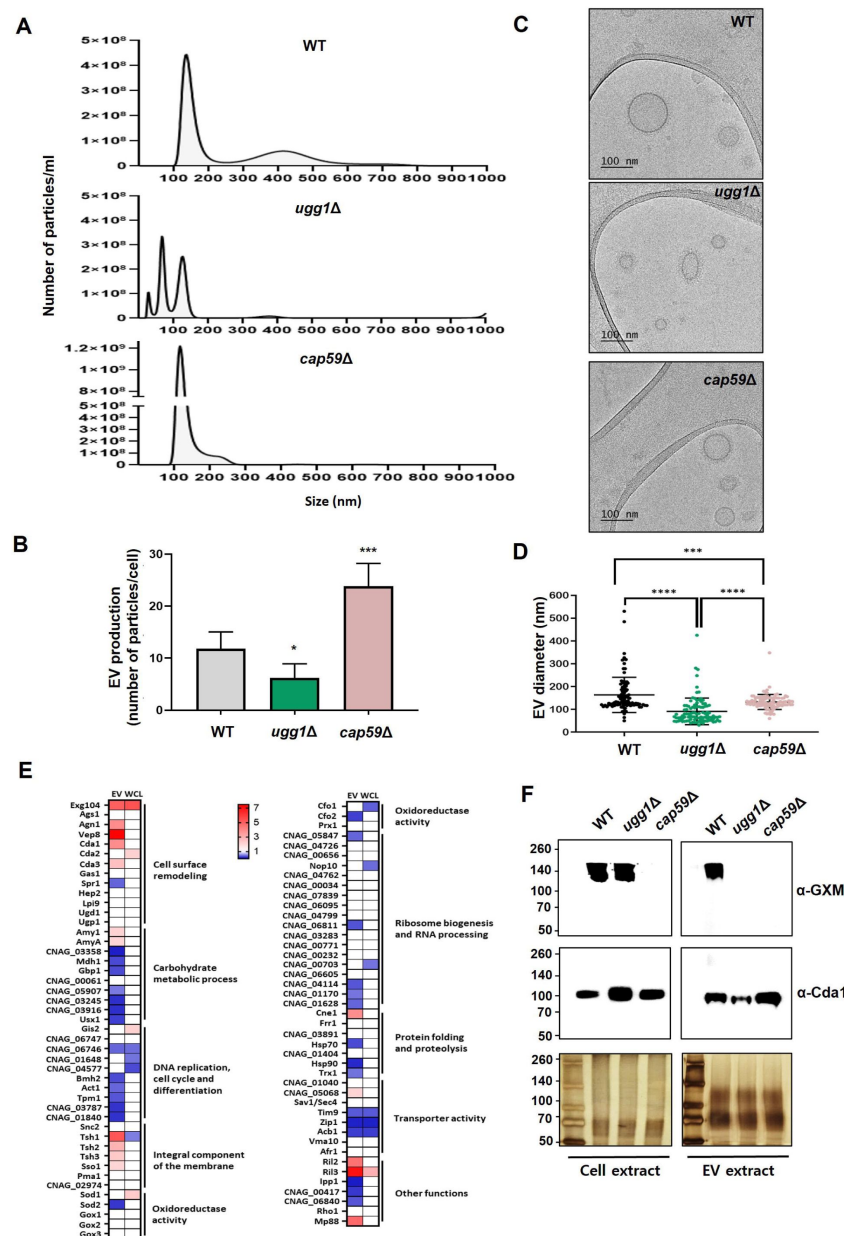


Figure 8.

Analysis of extracellular vesicles purified from WT, *ugg1Δ*, and *cap59Δ* cells.

(A, B) Nanoparticle tracking analysis (NTA) of EVs extracted from WT, *ugg1Δ*, and *cap59Δ* strains and quantification of total extracellular vesicle (EV) concentration per cell density. Quantitative measurements were derived from three independent experiments with standard deviations presented as error bars. Statistical significance: *** $P < 0.0005$, * $P < 0.05$. All statistical data were determined based on one-way ANOVA and Dunnett's post-hoc test. (C, D) Cryo-TEM imaging of purified EVs and comparative analysis of EV size in WT, *ugg1Δ*, and *cap59Δ* strains. Scale bar, 100 nm. The outer EV diameter of a total number of 100 EVs per strain, captured using cryo-TEM, were measured. (E) Heatmap representation of fold change between WT and *ugg1Δ* EV-associated proteins, commonly detected in this study and in previously reported EV proteome datasets (*ugg1Δ*/WT). Upregulated proteins in *ugg1Δ* are shown in red, whereas downregulated proteins are shown in blue. The proteome data of whole-cell lysates (WCL), generated from the cell pellets obtained after EV separation, were included for comparison. (F) Blotting analysis of GXM in the *C. neoformans* cells and EVs of WT, *ugg1Δ*, and *cap59Δ* strains. The 8 M urea extracts were obtained from EVs and cell pellets, from which EVs are generated. The urea extracts (5 μ g total proteins) were loaded on 8% SDS-polyacrylamide gel and subjected to silver staining or blotting analysis using the anti-GXM 18B7 (α -GXM) and anti-Cda1 (α -Cda1) antibodies, respectively. Left: total cell extract. Right: total EV extract.

respectively. We compared the EV protein content between WT and *ugg1Δ* identified in our study, focusing on overlapped proteins from the three EV proteomic data sets reported. Among 88 commonly reported EV-associated proteins, 29 were downregulated while 16 were overexpressed in *ugg1Δ* EVs compared to WT EVs (**Fig. 8E** [↗](#); Supplementary Table S4). The differential expression pattern of the EV-associated proteins is quite distinctive from that of WCL, revealing significantly detectable changes in EVs. Thus, the absence of functional Ugg1 not only caused morphological alterations and reduced EV numbers but also altered EV protein abundance in *C. neoformans*.

To further examine possible cargo loading defects, we investigated the presence of GXM in the *ugg1Δ* EVs, comparing GXM quantities in the total cell extracts from WT, *ugg1Δ*, and *cap59Δ* strains (**Fig. 8F** [↗](#)). Cda1 was used as the representative EV-associated protein as its presence was reported within the *C. neoformans* EV membrane, despite a substantial portion of Cda1 being secreted through the conventional secretion pathway (Rizzo et al., 2021 [↗](#)). GXM polysaccharides were not detected in the EVs released from *ugg1Δ*, although it was present in the whole-cell extract. Taken together with the reduced quantity of EV-associated proteins, the absence of GXM polysaccharides in *ugg1Δ* EVs strongly indicates that EV cargo loading is defective when the ERQC is dysfunctional in *C. neoformans*.

Discussion

As glycoproteins pass through the ER-Golgi secretory pathway, their *N*-glycans undergo extensive modifications in association with protein quality control. The ERQC system is highly conserved among eukaryotes but also diverged with distinct and species-specific features. The ERQC composition of *C. neoformans* is unique in that it possesses UGGT but lacks the three glucosyltransferases necessary for the addition of glucose residues to the core precursor *N*-glycans (Park et al., 2012 [↗](#)). Additionally, it lacks CRT, and carries multiple α 1,2-mannosidases. In this study, we investigated the molecular features and functions of the *N*-glycan-dependent ERQC in *C. neoformans*, which generates unique *N*-glycan precursors without glucose addition, and shorter in length than those of most eukaryotes. Our data strongly suggest that despite the incomplete composition of the ERQC components, the UGGT-centered ERQC plays pivotal roles in cellular fitness, and particularly in the EV-mediated extracellular transport, which is crucial for pathogenicity.

The virulence of the *C. neoformans* *ugg1Δ* mutant was almost abolished in mice (**Fig. 4D** [↗](#)), a phenotype consistent with the observed defects in key virulence determinants, such as the capsule and melanin, as well as poor growth at 37 °C (**Fig. 3B** [↗](#); **Fig. 4A** [↗](#) and **4B** [↗](#)). The *mns1Δ101Δ* mutant displayed intermediate phenotypes between those of the WT and *ugg1Δ*. Our data from the UPR induction analysis and comparative transcriptomic analysis strongly indicate that the ERQC mutation generates ER stress; this accounts for significant similarity of the defective phenotypes of the *C. neoformans* ERQC mutants to those reported in other mutants that exhibit defective protein folding in the ER, particularly in terms of increased stress sensitivity and decreased virulence. Connections between ER stress and thermotolerance have previously been established in *C. neoformans*, as growth at 37 °C requires key ER protein chaperones and protein processing machinery, including components of the UPR signaling pathway Ire1 and the ER stress-responsive transcription factor Hx11 (Cheon et al., 2011 [↗](#); Havel et al., 2011 [↗](#); Jung et al., 2013 [↗](#)). Other mutations causing defects in ER function in *C. neoformans*, such as mutants lacking *DNJ1* (an ER J-domain containing co-chaperone) and *CNE1* (an ER chaperone), resulted in growth inhibition. The *dnj1Δ* mutant strain displayed impaired elaboration of virulence factors, such as the exopolysaccharide capsule and extracellular urease activity, when cultured at human body temperature (Horianopoulos et al., 2021 [↗](#)). Altogether with our data from the ERQC mutants, these findings strongly support the notion that maintaining ER homeostasis is crucial for survival and virulence factor production at elevated temperatures.

It is quite notable that the *C. neoformans* *ugg1Δ* EVs exhibited defective loading of GXM and many protein cargo, alongside a decrease in the number and changes in the size distribution (Fig. 8). Combined with compromised cell fitness, these defects in EV biogenesis and cargo loading, even under normal growth conditions, likely contribute to the complete loss of virulence owing to the defective export of virulence factors in the *ugg1Δ* mutant. Interestingly, our comparative proteome analysis of culture supernatants (secretome) revealed that the *ugg1Δ* strain exhibited more pronounced defects in the secretion of proteins lacking signal peptides, which are often linked to unconventional secretion mechanisms, compared to those with canonical secretion signals (Supplementary Fig. S8; Supplementary Table S5). These findings further support the role of ERQC in modulating non-conventional secretion pathways.

It is intriguing how the absence of Ugg1, leading to ERQC defects, results in defective EV biogenesis and cargo loading in *C. neoformans*. In fungi, EV production and release occur through the maturation of endosomes into multivesicular bodies (MVBs), which are directed to the cell surface. Upon fusion with the plasma membrane, MVBs release vesicles into the extracellular environment, functioning as an unconventional secretion pathway (Oliveira et al., 2010). MVBs formation relies on the functionality of the endosomal sorting complex required for transport (ESCRT), a highly intricate pathway involving a series of finely regulated events (Henne et al., 2011). Deletion of several ESCRT complex-related genes involved in EV transport directly impacts the cryptococcal capsule, notably resulting in reduced capsule size. Specifically, the *C. neoformans* *vps27Δ* strain showed altered EV size distribution, reduced capsule dimensions, defects in laccase export to the cell wall, and poor extracellular export of urease (Park et al., 2020). Other regulators of unconventional secretion are also linked to EV biogenesis. For example, the Golgi reassembly and stacking proteins (GRASPs) regulate EV cargo and dimensions in *C. neoformans* (Peres de Silva et al., 2018). A *graspΔ* mutant strain produced EVs with dimensions that significantly differed from those produced by WT cells, along with attenuated virulence and abnormal RNA composition. Moreover, the GRASP protein Grasp homology 1 (Grh1) serves as a chaperone that directly influences EV cargo (Malhotra, 2013; Peres da Silva et al., 2018). Autophagy regulators, which participate in EV formation in other eukaryotes, also play a role in cryptococcal EV formation. An *atg7Δ* strain manifests hypovirulence, and EVs produced by this strain show slightly different RNA composition compared with that of the WT cells (Oliveira et al., 2016). Notably, our data of the transcriptome and proteome analysis did not show significant changes in the ESCRT complex and GRASPs.

In addition to protein folding and secretion, the ER is crucial for regulating lipid metabolism (Moncan et al., 2021). Consequently, ER stress significantly impacts lipid and sterol synthesis, although some of these mechanisms are yet to be clarified. LDs aid the UPR and ERAD in degrading misfolded proteins during ER stress in *S. cerevisiae* (Garcia et al., 2021). In the present study, we observed a drastic increase in LDs in the *ugg1Δ* mutant (Supplementary Fig. S6, A), consistent with the previous hypothesis that LDs help maintaining ER homeostasis. Furthermore, we observed increased number of vacuoles in *ugg1Δ* (Supplementary Fig. S6, B). This may thus indicate abnormal lipid homeostasis caused by the ERQC defects, which could, in turn, affect EV biogenesis. The importance of lipids and membrane regulators in proper EV formation and GXM export has been suggested in a previous study on the Apt1 flippase in *C. neoformans* (Rizzo et al., 2018). Additionally, ergosterol is essential for membrane fluidity, permeability, and protein transport (Ermakova and Zuev, 2017). Erg6 is involved in the ergosterol biosynthesis pathway and was identified as essential for the trans-Golgi network transport of proteins (Proszynski et al., 2005; Nes et al., 2009). In *C. neoformans*, the *erg6Δ* mutant released EVs with a significantly larger diameter than those of the WT, carrying increased levels of proteins and sterols, highlighting the role of ergosterol in cryptococcal EV biogenesis (Oliveira et al., 2020). Notably, the *ugg1Δ* mutant exhibited loss of microvesicles, which are derived from the plasma membrane (Fig. 8A). Our preliminary analysis data on the surface lipid fraction of the *ugg1Δ* mutant

indicated detectable alteration of sphingolipids and sterol profiles. Altogether, these findings suggest that ER stress caused by misfolded glycoprotein accumulation in ERQC-defective mutants may alter lipid composition, which could affect EV biogenesis.

A recent study using kidney cells presented strong evidence for a key role of ER stress in modulating EV biogenesis, demonstrating that ER stress decreases exosome production in mice (Fukuoka et al., 2023 [DOI](#)). They showed that T-cadherin is downregulated by ER stress through IRE1 α activation at mRNA and protein levels and that ER stress decreases EV production through adiponectin/T-cadherin-independent way, which may involve interferon pathway activation in mice. We similarly observed induced activity of the ER stress sensor Ire1 in *ugg1* Δ , even under normal growth conditions, suggesting a possible association between the Ire1-mediated UPR pathway and EV biogenesis (Fig. 9 [DOI](#)). Additionally, glycosylation regulates the biogenesis of small EVs and affects protein cargo loading efficiency in melanoma cells (Harada et al., 2020 [DOI](#)). This suggests that the altered *N*-glycosylation observed in *C. neoformans* *ugg1* Δ may influence cargo loading of certain EV-targeting glycoproteins. Despite their findings not being conducted in fungi, they provide valuable guides on the investigation of conserved mechanisms shared with *C. neoformans*. Further studies to investigate the mechanisms underlying the ERQC-mediated modulation of EV biogenesis and cargo loading in *C. neoformans* will provide insights into understanding the regulation, production, composition, and diversity of fungal EVs, enabling a better understanding of their biological function. Expanding our knowledge on pathogenic fungal EVs would pave the way for utilizing native or engineered EVs as promising candidates for therapeutic applications, including fungal infection diagnosis and vaccine development.

Materials and Methods

Strains, culture conditions, plasmids, and primers

The *C. neoformans* strains that were constructed and used in this study are listed in Supplementary Table S1A. The plasmids and primers used are listed in Supplementary Tables S1B and S1C, respectively. The construction of the deletion mutants is described in the Supplementary Information. The strains were typically cultured in YPD medium (1% yeast extract, 2% bacto peptone, and 2% glucose) at 30 °C with shaking (220 rpm). *C. neoformans* transformants were selected by culturing on YPD solid medium containing 100 μ g/ml nourseothricin (Jena Bioscience, Germany; indicated as YPD_{NAT}), YPD solid medium with 100 μ g/ml hygromycin B (Sigma-Aldrich, USA; indicated as YPD_{HyB}), or YPD solid medium with 100 μ g/ml G418 (Duchefa, Netherlands; indicated as YPD_{NEO}). For capsule induction, *C. neoformans* cells were cultured in liquid Sabouraud dextrose medium (Difco) at 30 °C for 16 h and incubated in 10% Sabouraud dextrose medium (pH 7.3) supplemented with 50 mM morpholinepropanesulfonic acid (MOPS) at 30 °C for 2 days. For melanin production analysis, the cells were spotted onto an L-DOPA agar plate (7.6 mM L-asparagine monohydrate, 5.6 mM glucose, 22 mM KH₂PO₄, 1 mM MgSO₄·7H₂O, 0.5 mM L-DOPA, 0.3 mM thiamine-HCl, and 20 nM biotin) and incubated at 30 °C or 37 °C for 2 days.

Transmission electron microscopy

C. neoformans cells were cultured either at an initial optical density of 600 nm (OD₆₀₀) of 0.2 at 30 °C in YPD medium until OD₆₀₀ reached 0.8, or for 2 days in 10% Sabouraud media at 30 °C. The cell pellets were washed twice in PBS and fixed for 12 h in 2% glutaraldehyde-2% paraformaldehyde in 0.1 M phosphate buffer (pH 7.4). The cells were washed in 0.1 M phosphate buffer and post-fixed with 1% OsO₄ in 0.1 M phosphate buffer for 2 h. The cells were dehydrated using an ascending ethanol series (50, 60, 70, 80, 90, 95, 100%) for 10 min each and infiltrated with propylene oxide for 10 min. The specimens were embedded with a Poly/Bed 812 kit (Polysciences Inc., USA) and polymerized in an electron microscope oven (TD-700, DOSAKA, Japan) at 65 °C for 12 h. The block was cut into 200-nm semi-thin sections with a diamond knife in the ultramicrotome and stained with toluidine blue for observation using optical microscopy. The region of interest was further

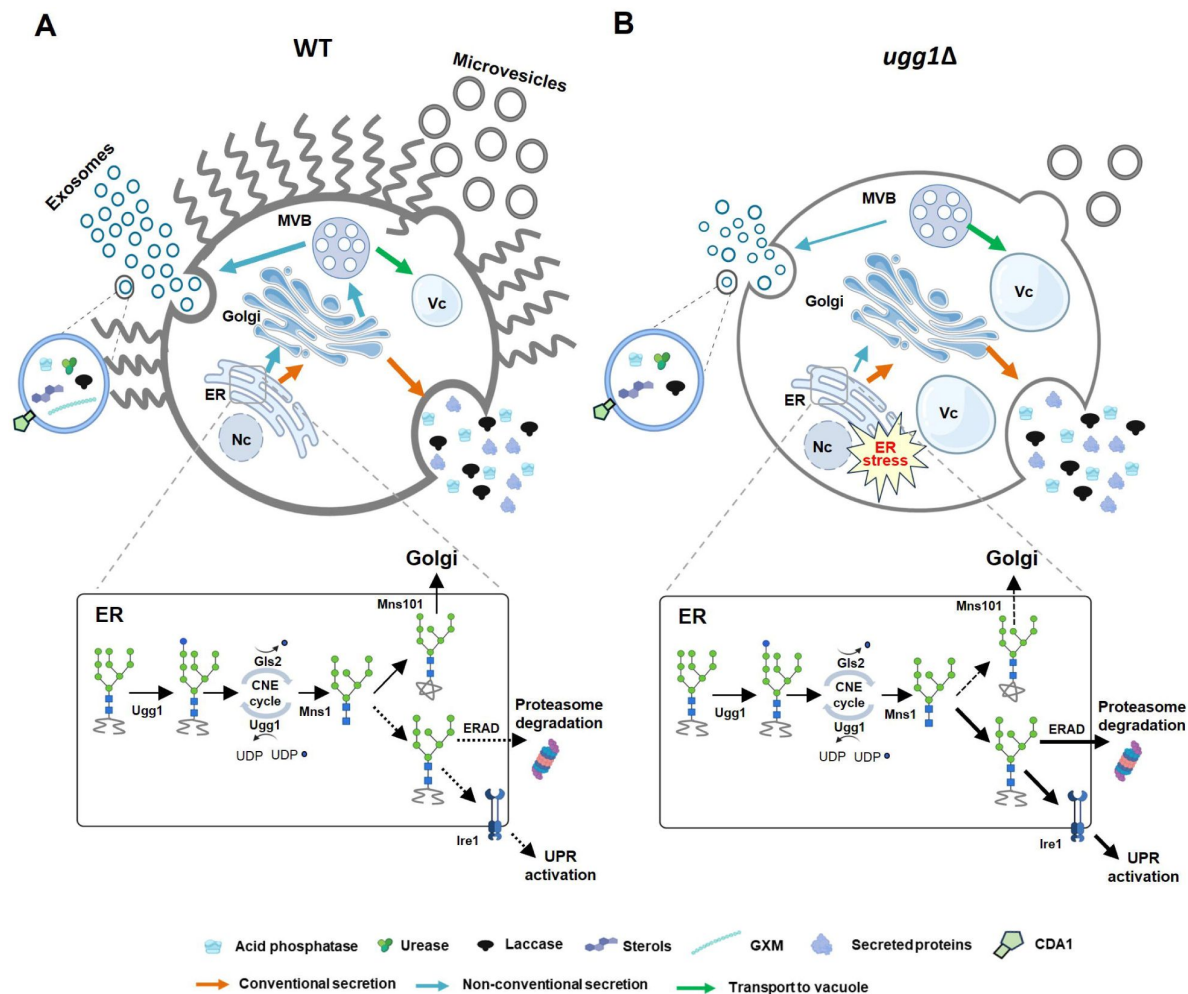


Figure 9.

Impact of ERQC disruption on glycoprotein folding and EV-mediated transport of virulence factors in *C. neoformans*.

(A) In WT strain, *C. neoformans* UGGT homolog, Ugg1, functions as a sensor for misfolded glycoproteins within the ER, playing a crucial role in protein quality control. Functional ERQC is essential not only for ensuring the proper folding of glycoproteins, which is critical for maintaining cellular fitness, but also for facilitating EV-mediated secretion of capsule polysaccharides and virulence-related enzymes necessary for pathogenicity. (B) In the UGGT-deficient strain (*ugg1Δ*), ER stress is increased because of misfolded protein accumulation within the ER lumen. This heightened stress leads to decreased cellular fitness, which negatively impacts EV biogenesis and cargo loading. Consequently, significant defects occur in EV-mediated transport, which ultimately leads to a complete loss of virulence. Nc: nucleus, Vc: vacuoles. This figure was partially created using BioRender (<https://BioRender.com/3gzzput>, <https://BioRender.com/bgibgte>).

cut into 80 nm thin sections using the ultramicrotome, placed on copper grids, double stained with 3% uranyl acetate for 30 min and 3% lead citrate for 7 min, and imaged using a transmission electron microscope (JEM-1011, JEOL, Tokyo, Japan) equipped with a Megaview III CCD camera (Soft imaging system-Germany) at the acceleration voltage of 80 kV.

Capsule transfer and shedding analysis

Capsule transfer assays were performed as described previously (Reese and Doering, 2003 [\[1\]](#)). Briefly, conditioned medium (CM) was prepared as a source of GXM by culturing the respective strains for 5 days in YPD medium, followed by filtering and storing the culture supernatant at 4 °C. The acceptor strains were cultured overnight at 30 °C in YPD medium. In total, 2×10^6 acceptor cells were incubated with 1 µl of CM for 1 h at room temperature under rotation (18 rpm) and washed twice with PBS. Capsule acquisition was visualized by incubating the cells with an anti-GXM (18B7) antibody conjugated with AlexaFluor 488 (ThermoFisher Scientific, USA) for 1 h at 37 °C, and observing under an Eclipse Ti-E fluorescence microscope (Nikon, Japan), equipped with a Nikon DS-Qi2 camera. The images were processed using the NIS-elements microscope imaging software (Nikon, Japan). Capsule shedding analysis was performed using a modified previously described protocol (Yoneda and Doering, 2008 [\[2\]](#)). The respective strains were cultured in 10% Sabouraud media for 2 days, and the culture supernatant was sterile filtered. Enzyme denaturation was performed by subjecting the filtrate to heating at 70 °C for 15 min, followed by centrifugation at 13,000 rpm for 3 min. Intracellular GXM analysis was performed by resuspending the pellets in TNE buffer (50 mM Tris-HCl [pH 7.5], 150 mM NaCl, 5 mM EDTA) with the same volume of glass beads (425–600 µm in diameter, Sigma-Aldrich, USA). The cells were disrupted four times for 15 s at 5,000 rpm using a Precellys 24 Tissue Homogenizer (Bertin Technologies, France), followed by centrifugation at 16,000 rpm for 5 min at 4 °C. Next, 10 µl of the supernatant of either secreted or intracellular polysaccharide was mixed with 6X loading dye and run on a 0.6% certified megabase agarose (Bio-rad, USA) gel in 0.5X TBE (44.5 mM Tris base, 44.5 mM boric acid, 1 mM EDTA [pH 8.0]) at 25 V for 16 h. The polysaccharides were transferred onto a nylon membrane using the southern blotting technique. The membrane was air-dried, blocked using 5% skim milk, and treated overnight with 2 µg/ml 18B7 antibody. After washing, the membrane was incubated with an anti-mouse peroxidase-conjugated secondary antibody and subjected to detection using chemiluminescence.

Detection of enzymatic activities in intracellular and secretory fractions

Biochemical enzymatic activities of acid phosphatase, urease, and laccase in the intracellular and secreted fractions were assayed spectrophotometrically, using previously described methods (Lev et al., 2014 [\[3\]](#); Fu et al., 2018 [\[4\]](#); de Sousa et al., 2022 [\[5\]](#)). Acid phosphatase activity was assayed by culturing cells in MM-KCL medium (0.5% KCl, 15 mM glucose, 10 mM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 13 mM glycine, and 3 µM thiamine) for 3 h at 30 °C. The culture supernatant and soluble cell lysate were allowed to react with 2.5 mM p-nitrophenyl phosphate (pNPP) for 30 min at 37 °C. Urease activity was determined after culturing the cells in Rapid Urea Broth (RUH broth) for 24 h at 30 °C and subjecting the culture supernatant and soluble cell lysate to incubation with phenol red. Laccase activity was assayed by culturing the cells in asparagine salts media (7.6 mM L-asparagine, 0.1% glucose, 22 mM KH_2PO_4 , 1 mM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.3 mM thiamine-HCl, and 20 nM biotin) for 48 h at 30 °C and allowing the supernatant and soluble cell lysate to react overnight with 10 mM L-DOPA. The reactions were quantified by measuring OD_{420} (acid phosphatase), OD_{570} (urease), or OD_{480} (laccase). Cellulase activity was determined by culturing the cells in cellulose media (0.1% NaNO_3 , 0.1% K_2HPO_4 , 0.1% KCl, 0.5% MgSO_4 , 0.5% yeast extract, 0.1% glucose, and 0.5% low viscosity carboxymethyl cellulose) for 24 h at 30 °C. The culture supernatants were concentrated using an Amicon tube (30 kDa cutoff, Sigma-Aldrich, USA). The concentrated supernatants and soluble cell lysates were allowed to react at 40 °C for 10 min with the substrate 4,6-O-(3-ketobutylidene)-4-nitrophenyl-β-D-cellopentaoside (BPNPG5), which was provided in the cellulase assay kit (CellG5 Method, Megazyme, Ireland). The reaction was terminated by adding 2% [w/v] Tris buffer (pH 10),

and the absorbance of 4-nitrophenol was measured at 400 nm. Cellulase activity (CellG5 Units/ml) was calculated as indicated in the cellulase assay kit. α -amylase activity was measured using the α -Amylase Activity Colorimetric Assay Kit (Biovision Technologies, USA) according to the manufacturer's instructions. Yeast cells were cultured overnight in YPD medium at 30 °C, and the culture supernatants were concentrated using an Amicon tube (30 kDa cutoff, Sigma-Aldrich, USA). The concentrated supernatants and soluble cell lysates were incubated for 1 h at 25 °C in assay buffer with the substrate ethylidene-pNP-G7, which was provided in the kit. OD₄₀₅ was measured. All the activity analysis results were normalized according to cell density (OD₆₀₀).

Purification, nanoparticle tracking analysis (NTA), and Cryo-TEM imaging of extracellular vesicles

EV purification was performed according to a previously published protocol (Reis et al., 2019) with modifications. One loop of cells was inoculated into 10 ml of liquid YPD and incubated at 30 °C for 24 h with shaking (220 rpm). The cells were washed twice with 10 ml of PBS, counted, and diluted in PBS to a density of 3.5×10^7 cells/ml. Aliquots of cell suspension (300 μ l) were spread onto synthetic dextrose (SD) solid medium plates and incubated for 24 h at 30 °C. The cells were carefully recovered from each plate using an inoculation loop, gently resuspended in 30 ml PBS, and pelleted through centrifugation at 4,000 rpm for 10 min at 4 °C. The supernatant was collected and centrifuged again at 15,000 $\times g$ for 15 min at 4 °C. Then, the supernatant was filtered through 0.45- μ m syringe filters and ultracentrifuged at 100,000 $\times g$ for 1 h at 4 °C (MLA-50 fixed angle rotor, Beckman Coulter, Germany). The supernatant was discarded, and the EV pellets were collected and resuspended in 1 ml of PBS for immediate use or stored at -80 °C for further experiments. The samples were diluted 100-fold, and EV sizes were measured using an NTA instrument (NFEC-2024-03-295455, Nanosight Pro, Malvern Panalytical, Netherlands) coupled to a 532-nm laser (Malvern Panalytical, Netherlands), SCMOS camera (Hamamatsu Photonics, Japan), and syringe pump (Malvern Panalytical, Netherlands). The data were analyzed using the NS Xplorer software (v1.1.0.6, Malvern Panalytical, Netherlands).

Cryo-TEM imaging was performed by loading purified EVs (3 μ l) onto a lacey carbon grid (Lacey Carbon, 300mesh Cu, Ted Pella Inc., USA), which was glow discharged at 15 mA for 60 s. The sample-loaded grid was blotted for 3 s at 15 °C and 100% humidity and immediately plunge-frozen in liquid ethane. The process was performed by Vitrobot Maek IV (ThermoFisher Scientific, USA, SNU, CMCI). The frozen grids were imaged using TEM (JEM-2100F, JEOL, Japan); the temperature of the grid was maintained at approximately -180 °C at an acceleration voltage of 200 keV. The images were recorded using an ultrascan 1000 electron detector.

Proteomic analysis based on mass spectrometry and data processing

The samples for proteomic analysis were prepared as described in Supplementary Information. The labeled peptides (total, 100 μ g proteins) were combined prior to offline basic reverse-phase liquid chromatographic (bRPLC) fractionation. Linear gradient was performed using buffer A (10 mM TEAB in water) and buffer B (10 mM TEAB in 90% acetonitrile), and 10 fractions were analyzed in total using an LC-MS/MS system. The samples were dissolved in 0.1% formic acid using an UltiMate 3000 RSLCnano system and analyzed using an Orbitrap Eclipse Tribrid mass spectrometer (Thermo Fisher Scientific, USA). All MS raw files were converted into mzML and ms2 file formats using the MSConvert (version 3.0.20033) software. The *C. neoformans* proteome was determined by downloading a protein FASTA file from Uniprot (<http://uniprot.org>), which included 7,492 reviewed (Swiss-Prot) and unreviewed (TrEMBL) proteins entries. A proteome search database with reversed sequence and contaminants in the Integrated Proteomics Pipeline (IP2, version 5.1.2, Integrated Proteomics Applications Inc., San Diego, CA) was generated. The proteome search from 20 ms2 files was performed with IP2 and its following parameters, followed by evaluation for a false discovery rate (FDR) using IP2 and ProteinInferencer (version 1.0,

Integrated Proteomics Applications Inc., USA). Protein quantification and statistical analysis for discovery of differentially expressed proteins (DEPs) was performed using the ms2 files with tandem mass tag (TMT) reporter ions using an in-house program coded using Python 3.8, where t-test and Pearson's correlation analysis between comparison samples was performed using the scikit-learn (version 0.23.2), Scipy (version 1.6.0), and statsmodels (0.12.1) Python libraries.

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Additional information

Data availability

The raw RNA sequencing data have been submitted to the NCBI GEO database under accession no. GSE254772.

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Author Contributions

Catia Mota: Investigation, Construction of mutants, Performing all virulence analysis, Data curation, and Writing—original draft; Kiseung Kim: Construction of mutants and Performing phenotype analysis; Ye Ji Son: Transcriptome analysis and Validation; Eun Jung Thak and Su-Bin Lee: Glycan analysis; Ju-El Kim and Jeong-Kee Yoon: NTA analysis; Min-Ho Kang: Cryo-TEM analysis; Heeyoun Hwang: Proteome analysis; Yong-Sun Bahn and J. Andrew Alspaugh: Supervision, Data curation, Writing—review, and editing; Hyun Ah Kang: Conceptualization, Project administration, Supervision, Data curation, Writing—original draft, review, and editing.

Additional files

Supplementary material and methods. (a) Construction of *C. neoformans* *ugg1Δ* and UGG1 complementation strains. (b) Construction of *C. neoformans* *mns1Δ*, *mns101Δ*, *mns1Δ101Δ*, and respective complementation strains. (c) Construction of *C. neoformans* *mnl1Δ*, *mnl2Δ*, and *mnl1Δmnl2Δ* strains. (d) Construction and localization analysis of *C. neoformans* GFP-Ugg1, Mns1-GFP, and Mns101-GFP fusion proteins. (e) HPLC and MALDI-TOF-based N-glycan structure analysis. (f) Animal study and in vitro survival analysis. (g) Subcellular fractionation and western blot analysis of Cda1. (h) RNA preparation, qRT-PCR, and RNA-sequencing. (i) Sample preparation for proteomic analysis. [↗](#)

Supplementary figures. (a) Figure S1. Evolutionary conservation and divergence of endoplasmic reticulum quality control (ERQC)-related gene homologues predicted in different eukaryote organisms. (b) Figure S2. Domain analysis of putative *C. neoformans* ERQC-associated proteins. (c) Figure S3. Phenotype characterization of *C. neoformans* α1,2-mannosidases. (d) Figure S4. Subcellular localization of GFP-tagged Ugg1, Mns1, and Mns101. (e) Figure S5. Fungal burden of mice infected with *C. neoformans* ERQC-related mutant cells. (f) Figure S6. Lipid and vacuole staining of WT and *ugg1Δ* strains. (g) Figure S7.

Comparative analysis of whole-cell lysate (WCL) and EV proteomes between WT and *ugg1Δ* strains. (h) Figure S8. Comparative analysis of whole-cell lysate (WCL) proteome and secretome between WT and *ugg1Δ* strains. [↗](#)

Table S1. (A) List of strains used in this study. (B) List of plasmids used in this study. (C) List of primers used in this study. [↗](#)

Table S2. (A) Up-regulated (> 2-fold) *Cryptococcus neoformans* genes in *ugg1Δ* than that of wild type (WT) cultivated in yeast extract peptone dextrose (YPD) at 30 °C. (B) Down-regulated (> 2-fold) *Cryptococcus neoformans* genes in *ugg1Δ* than that of wild type (WT) cultivated in YPD at 30 °C. [↗](#)

Table S3. Excel file containing the list of proteins identified in the proteomic analysis of the wild type (WT) and *ugg1Δ* extracellular vesicles (EVs) and whole-cell lysates (WCL). [↗](#)

Table S4. Representative extracellular vesicle (EV)-associated proteins commonly detected in this study and previously reported *Cryptococcus neoformans* EV proteome data sets. [↗](#)

Table S5. Excel file containing the list of proteins identified in the whole-cell lysate (WCL) proteome and secretome analysis in the wild type (WT) and *ugg1Δ* strains. [↗](#)

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Reviewer #2 (Public review):

Summary:

This study investigates the molecular function of the N-glycan-dependent endoplasmic reticulum protein quality control system (ERQC) in *Cryptococcus neoformans* and correlates this pathway with key features of *C. neoformans* virulence, especially those mediated by extracellular vesicle transport. The findings provide valuable insights into the connection between this pathway and the biogenesis of *C. neoformans* extracellular vesicles.

Strengths:

The strength of this study lies primarily in the careful selection of appropriate and current methodologies, which provide a solid foundation for the authors' results and conclusions across all presented data. All experiments are supported by well-designed and established controls in the study of *C. neoformans*, further strengthening the validity of the results and conclusions drawn from them. The study presents novel data on this important pathway in *C. neoformans*, establishing its connection with *C. neoformans* virulence. Interestingly, the findings led the authors to understand the relationship between this pathway and the transport of key fungal virulence factors via extracellular vesicles. This was demonstrated in the study, paving the way for a deeper understanding of extracellular vesicle biogenesis—a field still filled with gaps but one that this study contributes to with solid data, helping to clarify aspects of this process.

<https://doi.org/10.7554/eLife.103729.2.sa2>

Reviewer #3 (Public review):

Summary:

Cryptococcus neoformans is a global critical threat pathogen and the manuscript by Mota et al demonstrates that the pathogen's N-glycan-dependent protein quality control system regulates the capacity of the fungus to cause disease. The system ensures that glycoproteins are folded correctly. The system is involved in fitness and virulence of the fungus by regulating aspects of cellular robustness and the trafficking of virulence-associated compounds outside of the cell via transport in extracellular vesicles.

Strengths:

The investigators use multiple modalities to demonstrate that the system is involved in cryptococcal pathogenesis. The investigators generated mutant *C. neoformans* to explore the role of genes involved in the protein folding system. Basic microbiology, genetic analyses,

proteomics, fluorescence and transmission microscopy, nanotracking analyses, and murine studies were performed. The validity of the findings are thus very high. Hypotheses are robustly demonstrated.

<https://doi.org/10.7554/eLife.103729.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Recommendations for the authors):

Major comments

(1) The section on page 20 describing the proteomic analysis of EVs is poorly written and confusing, with a lot of data in the supplement. It is not clear what the proteomics data actually means.

We appreciate your feedback on the clarity of the proteomic analysis section. We have rewritten the section on page 20 with more detained information to provide a clearer explanation of the proteomics data and its biological significance. Additionally, we have incorporated a comparative analysis of the EV and total cell lysate proteomes (Fig. 8E, Supplementary Fig. S7A, Supplementary Tables 3 and 4) for supplemental data interpretation.

(2) The order of the data could be improved.

We appreciate your feedback regarding the data organization. We have reorganized the order and position of some data in a more structured and coherent manner, as suggested by the reviewers.

- Reorganization of the qPCR data (previously Fig. 1C) as Fig. 3A
- Removal of the data on the growth analysis on raffinose media (previously Fig. 7H).
- Reorganization of the spotting data of the double mutant (previously Fig 3B) to Supplementary Fig. S3B
- Reorganization of the subcellular localization data (previously Fig 3E) to Supplementary Fig. S4A

(3) The discussion is repetitive with the introduction and merely summarizes the results and speculates on the mechanism of how the absence of UGGT, leading to ERQC defects, results in defective EV biogenesis/cargo loading in C. neoformans.

We removed several repetitive sentences in the discussion and provided additional information on proteome analysis.

Other questions and comments

(1) Instead of comprehensively analyzing EVs from the UGG1 mutant, a more informative approach to better understanding how defects in N-linked glycosylation impact secretion, would be to do a proteomic analysis on the total secretions (including beta glucanase-treated cells to release classically secreted proteins from the cell wall) and EVs.

We agree that a comprehensive proteomic analysis of total secretions and classically secreted proteins would provide deeper insights into how defects in N-glycosylation impact secretion

in *C. neoformans*. To address this concern, we performed an additional set of proteomic analyses, the proteome profiles of total cell lysates and the secretome of *C. neoformans* cultivated in SD broth and presented the results as Supplementary Table S5 and Supplementary Fig. S7B. These additional analyses provide further insights into the impact of *UGG1* deletion on both conventional and unconventional secretion pathways, supporting a more pronounced effect of the *UGG1* defect on EV-mediated trafficking. The discussion has been updated accordingly (Page 22, lines 509-514).

(2) The melanization defect in Ugg1 mutant is not strong. Could the reduction be due to partially compromised Ugg1 mutant growth at 30°C as indicated in the spot tests. Were photos of the spot dilution assays taken at 1 and 2 days to investigate slower growth? Or alternatively were growth curves taken in a liquid culture?

For accuracy of melanin synthesis defect, in addition to analysis on L-DOPA plates, we had assessed melanin production in liquid L-DOPA medium following a 3-day incubation, and the melanin production in liquid media was normalized by cell density (OD_{600}). The data on normalized melanin production is now included as Fig. 4B in the revised manuscript. The defective laccase activity in the *ugg1Δ* mutant (Fig. 7C) further corroborates our melanization assay results, which is additionally mentioned in the text (Page 18, lines 393-395).

(3) Is it accurate to say that some virulence factors (i.e. melanin, capsule and phosphatases) are predominantly trafficked through EV's in C. neoformans? Have studies been done to determine the proportion of virulence factors trafficked via EV's versus traditional secretion?

We thank you for the thoughtful comments. Some virulence factors, such as urease, melanin and capsule polysaccharides, lack a signal peptide required for targeting for the conventional ER/Golgi secretion pathway. It is generally assumed that the trafficking of these factors in *C. neoformans* is predominantly mediated by non-conventional secretion via EVs. Additionally, even some virulence factors with signal peptides, such as laccase and phosphatases, are also transported via EVs besides the conventional secretion. The quantitative analysis to compare the proportion of virulence factors secretion via EVs versus the conventional pathway has not been yet reported, despite that genetic evidence suggests that conventional secretion also plays a significant role in the export of capsule polysaccharides. Thus, we were also careful not to highlight EV as the main route of virulence factors in the manuscript.

(4) There is insufficient background in the introduction linking what is known about the ERQC process to secretion in general. The topic changes from the ERQC process to fungal virulence factor, with a primary focus on non-classical (EV-based) secretion. Classical secretion should also be discussed without assuming that non classical (EV) secretion is the major pathway contributing to fungal virulence.

We appreciate your insightful comments highlighting the need for more background on the ERQC process and its relationship with secretion. To address the reviewer's concerns, we have added sentences to describe the key roles of ERQC in conventional protein secretion in the Introduction (Page 5, lines 102-106).

(5) Figure 1A. What does the blue filled circle with the red outline signify? Fig 1 A legend is not well explained. A summary using material provided in the intro/discussion should be included to briefly explain the process and the differences between fungal species. Please also be aware that the intro starts describing the human ERQC process and then switches to what happens in S. cerevisiae.

We have revised Figure 1A by removing the red circle and updated the figure legend in the revised manuscript to include more detailed information about the ERQC differences across higher eukaryotes and fungal species.

(6) Figure 2A. There are no units on the Y-axis. Presumably, the scale is the same for all 3 strains.

Thank you for your comments. The Y-axis is the same for all three strains and, as in Fig. 2C, and represents the relative fluorescence intensity obtained from the HPLC analysis. We added the units on the Y-axis in Fig. 2A.

(7) If Mnl1 and 2 have proposed roles in proteasomal degradation, wouldn't they be expected to have ER retention signals, like Ugg1?

We appreciate your valuable insights regarding the absence of ER retention signals in Mnl1 and Mnl2. Previous studies have shown that *Saccharomyces cerevisiae* Mnl1/Htm1 does not possess canonical KDEL/HDEL-like ER retention signals. Instead, its retention in the ER lumen is facilitated through its interaction with protein disulfide isomerase Pdi1, which contains an HDEL sequence (Gauss et al. 2011). Thus, it is expected that non-canonical retention mechanisms—such as interactions with other ER proteins—could contribute to the retention of Mnl1 and Mnl2 within the ER. We added this information to the revised manuscript (Page 8, lines 154-159).

(8) Figure 1 C qPCR showing change in mRNA in response to ER stress should not be grouped in this figure. It could be standalone or discussed when the spot dilution assays are performed. Anyway, spots tests are more convincing of a role in stress response than qPCR as the *ugg1* mutant is sensitive to tunicamycin, DTT and cell wall stressing agents.

As suggested by the reviewer, we have reorganized the qPCR data as a part of Figure 3 (Figure 3A) in the revised manuscript.

(9) It is odd that *mns1/101* mutants are not sensitive to ER and CW stress given their proposed differing location/function in the pathway (Figure 1) determined from the N-linked profiling. Any explanation? Could there be redundancy?

We appreciate the reviewer's observation regarding the lack of ER and CW stress sensitivity in the *mns1Δ* and *mns101Δ* mutants, despite their proposed roles in N-glycan processing. We had previously reported that the *C. neoformans alg3Δ* mutant, lacking a critical enzyme responsible for the synthesis of Dol-PP-Man₆GlcNAc₂ in the N-glycosylation pathway, exhibited clearly impaired N-glycan elongation, but showed no detectable growth defects even under stress conditions *in vitro*. However, *alg3Δ* is avirulent in *in vivo* pathogenicity (Thak et al., 2020). Similarly, the *mns1Δ101Δ* double mutant shows glycan-processing defects that do not compromise cellular fitness under stress conditions but result in attenuated virulence in animal models. These findings suggest that some glycosylation-related defects may impact more severely *in vivo* pathogenicity rather than *in vitro* stress sensitivity.

(10) Although the Silver-stained gels of the *ugg1* mutant are not particularly informative, why weren't they (and Con A blots) performed for the other mutants?

The overall decrease of hypermannosylated glycans observed in the *ugg1Δ* mutant allowed us to detect clear alterations in protein glycosylation patterns in the lectin blot using *Galanthus nivalis* agglutinin, which recognizes terminal α1,2-, α1,3-, and α1,6-linked mannose residues. In contrast, the limited changes of a few glycan species in other mutants, including *mns1Δ*,

mns101Δ, and *mns1Δ101Δ*, are relatively subtle to be detected in the lectin blot, due to only minor differences in the average lengths of their *N*-glycans compared to the WT. Therefore, we presented the lectin blotting data only for the *ugg1Δ* mutant.

(11) *If there is ER stress under normal conditions in the Ugg1 mutant then technically this mutant should be growing more slowly under normal conditions. This is difficult to predict in a spot dilution assay where growth is only visualized at day three when any growth defect may have been corrected. The slower growth rather than the reduced secretion of GXM specifically is therefore more likely to be responsible for the reduced virulence.*

We appreciate the reviewer's insightful comment regarding the interplay between ER stress, growth defects, and virulence attenuation in the *ugg1Δ* mutant. While retarded growth in *C. neoformans* is often associated with reduced virulence, there are a few exceptions. For instance, disruptions in cell cycle progression in *C. neoformans* have been reported to result in larger capsule sizes, which rather enhance *in vivo* virulence when analyzed in *Galleria mellonella* infection models (García-Rodas et al., 2014). This highlights that growth defect alone is not sufficient for virulence attenuation. In the case of the *ugg1Δ* mutant, we speculate that the almost complete loss of virulence is attributed not only to its growth retardation but also to its impaired secretion of key virulence factors, including the polysaccharide capsule.

(12) *The rationale for using leucine analogue 5',5',5'-trifluoroleucine (TFL), in a growth assay (Fig. 3C) to determine whether the defective ugg1Δ phenotypes are induced by ER stress caused by misfolded protein accumulation is not explained.*

The leucine analogue 5',5',5'-trifluoroleucine (TFL) can be incorporated into newly synthesized proteins, disrupting normal folding and thus leading to the generation of misfolded proteins (Trotter et al., 2002; Cowie et al., 1959). In the context of a defective ERQC pathway, these misfolded proteins cannot be adequately repaired, resulting in their accumulation and triggering ER stress. Excessive ER stress may ultimately inhibit cell growth in the presence of TFL. This explanation has been incorporated into the revised manuscript (Page 11, lines 236–241).

(13) *I would argue that only the Ugg1 and double Mns mutant were defective in virulence. For the single mutants, it looks like no difference was found relative to WT. The longer median survival of these mutants (if significant) is most likely due to poor infection technique.*

We agree with the reviewer's opinion that the *mns1Δ* and *mns101Δ* single mutants have no significant difference in *in vivo* virulence compared to the WT strain, unlike the *mns1Δ101Δ* double mutant which showed significant attenuated virulence. We had previously addressed that in the manuscript (Page 13, lines 267–269).

(14) *The authors conclude that the ugg1Δ strain specifically is impaired in extracellular secretion of capsular polysaccharides but is this via classical (SAV1) secretion or EVs?*

In addition to EV-mediated transport, capsular polysaccharide secretion can occur via the Sav1 (Sec4p)-mediated classical secretion pathway. However, our proteome data of total cell lysates indicated that the protein levels of Sav1 were comparable between the WT and *ugg1Δ* strains, suggesting that Sav1p function itself might not be impaired. Given that the *ugg1Δ* mutant exhibits altered vesicular structures (Supplementary Fig. S6) and loss of microvesicles (Fig. 8A), we speculate that a defect might occur at a post-Sav1p step, such as vesicle fusion with the plasma membrane, likely contributing to the complete defect in secretion of capsular polysaccharides in the *ugg1Δ* strain, in which EV biogenesis and defective cargo loading are

severely impaired, producing EVs that lack capsular polysaccharides (Figure 8F). However, further studies should be carried out to define the contribution of SAV1 to the secretion of capsular polysaccharides in the *ugg1Δ* strain.

(15) The rationale for doing 7 H is very confusing.

The experiment assessing raffinose utilization as a carbon source was inspired by the previous work of Garcia-Rivera et al., reporting that the *cap59Δ* mutant is unable to utilize raffinose due to a defect in the secretion of raffinose-hydrolyzing enzymes. As another way to investigate potential defects in the conventional secretion pathway, we investigated the growth of the *ugg1Δ* mutant in the presence of raffinose. Due to our extensive data length, we have decided to remove this complementary data from the manuscript.

(16) It is speculated in the discussion that ER stress impacts lipid/sterol synthesis and that LDs (lipid droplets?) aid the UPR and ERAD in degrading misfolded proteins during ER stress in S. cerevisiae. The authors mention that they observed a drastic increase in LDs in the ugg1Δ mutant. Where is this data? Even with the data, this is all speculation. The authors also speculate that increased numbers of vacuoles in ugg1 (where is the data?) could be the cause of the altered vesicular structures observed in the mutants, which may indicate abnormal lipid homeostasis caused by the ERQC defects, which could, in turn, affect EV biogenesis. Again, this is speculative.

The data on lipid droplets (LDs) and vacuole staining are presented in Supplementary Figure S6, showing a drastic increase in LDs and an increased in vacuolar size in the *ugg1Δ* mutant compared to the wild-type strain, especially in capsule-inducing conditions. In addition to such changes in vesicular structures, our preliminary data on sphingolipids and sterol analysis in the surface lipid fraction of the *ugg1Δ* mutant led us to propose the hypothesis that ERQC defects may impact lipid metabolism, which in turn could influence EV biogenesis and membrane properties. It is expected that these findings would provide a strong foundation for future studies exploring the link between ERQC, lipid homeostasis, and EV biogenesis. We have revised our speculation on the association of abnormal lipid homeostasis, caused by ERQC, with EV biogenesis more appropriately by adding the information on our preliminary data of lipid profiles and mentioning that the *ugg1Δ* mutant lacks microvesicles, which are derived from the plasma membrane (Page 24, lines 554-559).

Reviewer #2 (Recommendations for the authors):

(1) My suggestions for the authors are the same as those presented in the public review: (1) reducing the text in certain sections of the paper to improve readability for the audience, and (2) reconsidering the figures to reduce the amount of information in each one, moving some of the content to the supplementary material.

We thank the reviewer for their constructive suggestions regarding the organization and readability of the manuscript. As suggested, we addressed your concerns as follows:

- (1) Reducing the text in the Introduction, Results, and Discussion sections by removing repetitive statements and simplifying complex descriptions where possible.
- (2) Changing the presentation of figures: we have also reorganized the presentation of some data by moving non-essential data to the supplementary material. The updated figures and supplementary materials have been clearly referenced in the text to guide readers.
- (3) Reorganization of materials and methods: some parts of methods were moved to Supplementary Information
- (4) Removal of Figure 7H and the sentences describing the result

More detailed explanations on the reduction and reorganization are also described in the response to the major comments (2) and (3) made by Reviewer #1.

(2) Figure 3, for example, shows no difference in fungal growth under different cultivation conditions. This information is valuable but could be mentioned in the text, with the image provided as supplementary material, focusing the figure only on images that show significant growth differences among the strains. I suggest a similar approach for other figures so that the authors can include only the most relevant results in the main body of the article and move some figures to the supplementary materials.

For Fig. 3, the spotting data of the double mutant (previously Fig. 3B) is now presented in the supplementary information (Supplementary Fig. S3B). Additionally, the subcellular localization data (previously Fig 3E) was also moved to the supplementary material (Supplementary Fig. S4A).

Reviewer #3 (Recommendations for the authors):

(1) Line 43 "EV-mediated transport of virulence bags" doesn't make sense. EVs have been described as "virulence bags" (and are in this work later in the introduction) but this should here be "transport of virulence factors" or "compounds associated with virulence" but only if you have confirmed that the "cargo" is consistent with this- which is not evident in the abstract.

Thank you for your insightful comment. We have revised this to "EV-mediated transport of virulence factors" in line with your suggestion.

(2) Line 49 "secretory pathway" - is there not more than one secretion pathway?

Thank you for pointing this out. The term "secretory pathway" has been updated to "secretory pathways" to acknowledge the presence of both conventional and unconventional secretion mechanisms.

(3) Line 53 "recognizes folding defects, repairs them, and ensures the translocation of irreparable misfolded proteins" should be "recognizes folding defects and repairs them or ensures the translocation of irreparable misfolded proteins."

Thank you for pointing this out. We have revised the sentence as you suggested.

(4) Lines 88-90 ALG needs to be written out the first time - Asn-linked glycans. Also, consider adding that ALG genes are present in most eukaryotes as it is unclear what you are comparing C. neoformans to.

Thank you for your helpful comment. We have revised the text to write out "ALG" as "Asn-linked glycosylation" and added the sentence "ALG genes are evolutionary conserved in most eukaryotes" in the revised manuscript (Page 4, line 84).

(5) Line 99 Cryptococcus has already been abbreviated to C. so don't write it out again.

We have corrected "Cryptococcus" to "C." throughout the manuscript after its first mention.

(6) Line 152- tunicamycin and DTT are not described yet, which may make it challenging for some readers to understand what these drugs are doing/why they were used. What is on lines 156 and 157 for these drugs should go up with the first mention of these drugs.

Thank you for your helpful suggestion. We have revised the manuscript to include the descriptions and purpose of using tunicamycin (TM) and dithiothreitol (DTT) immediately following their first mention, as recommended (Page 10, lines 208-210).

(7) *The text for Figure 1 C is inaccurate. High temperature also induced KAR2, as noted above, but inaccurately stated in line 160. There is no comment on the significant UGG1 increase with tunicamycin or that KAR2 was highest in this condition.*

Thank you for your thoughtful comment. We have better clarified the significant increase of *UGG1* expression following tunicamycin treatment and *KAR2* induction upon heat stress in the revised manuscript (Page 10, lines 216-217). Please note that Fig. 1C was revised and is now referred to as Fig. 3A.

(8) *Figure 2B is not well explored/explained. There appears to be more protein in the mutant, including of higher weight in the intracellular compartment. It is difficult to ascertain if there is more too in the secretion phase with this gel. The methods do not specifically describe the concentration of protein added - just volume. Is what we are seeing a loading issue vs real differences?*

Thank you for your insightful comments regarding Figure 2B. We added information on amounts of protein (30 µg per lane) in the legend of Figure 2B.

The main purpose of Fig. 2B is to examine the altered glycosylation pattern of ERQC by detecting glycoproteins using the *Galanthus nivalis* agglutinin, which specifically bind terminal α1,2-, α1,3-, and α1,6-linked mannose residues. The result of lectin blotting indicated that glycoproteins are more abundantly detected in the secretion fraction compared to in the soluble intracellular fraction, consistent with the general notion that more than 50% of secretory proteins are glycoproteins. Also, the more abundant proteins with decreased molecular weight in the secretion fraction of *ugg1Δ* mutant supported the *N*-glycan profiles with decreased hypermannosylation in *ugg1Δ* mutant. We added the purpose and more detailed interpretation on Figure 2B in the revised manuscript (Page 9, lines 174-179).

(9) *Line 242 "melanin pigment" is redundant as melanin is a pigment.*

We thank the reviewer for pointing out the redundancy in the phrase. We revised the text to simply state "melanin".

(10) *Line 250 drops "completely" especially as the mutant did colonize the lungs of mice.*

To avoid any possible misleading, we removed the term "completely" in the revised manuscript.

(11) *Line 275- need to reference 18B7 as it is first introduced here.*

We added the reference on the antibody 18B7 in the revised manuscript.

(12) *Line 308- there are specific techniques to measure GXM size that could validate or refute the statement on "incomplete" polysaccharides. For example, DOI:10.1128/EC.00268-09.*

We appreciated the valuable suggestion on specific techniques to measure GXM size, which will be one of key experiments in our future study. In the revised manuscript we cited the suggested reference to indicate the need for validation of our statement (Page 14, lines 316-318).

(13) Line 496 "mammals" - why is this used when the study is on a fungus, not a mammal? The structure of the first 2 paragraphs can be clearer to focus more on fungal biology.

We have compared both mammals and fungi to emphasize that the ERQC system is conserved among eukaryotes but diverged with a few species-specific features. This comparison is relevant in the context of understanding the evolutionary unique features of ERQC pathways in *C. neoformans*. We modified the first 2 paragraphs to clarify the main issue of our present study (Page 21, lines 472-483).

(14) Line 525- the *ugg* mutant was not avirulent as CFU was present and histopathology in the supplementary figures shows the tissue with *ugg1* deletion was not normal (although the images are not especially easy to review). Yes, the mutant did not kill under your test conditions, but it was not avirulent (incapable of causing disease). Significantly attenuated or other descriptors should be utilized. Line 548 is also thus incorrect "complete loss of virulence").

We appreciate the reviewer's concern regarding the description of the *ugg1Δ* mutant as avirulent. We agree that the use of merely "avirulent" may not fully capture the observed phenotypes in the CFU and histopathological data, since we cannot exclude the possibility that the *ugg1Δ* mutant retains the ability to establish an infection. Thus, we have revised the text by describing the *ugg1Δ* mutant as "almost avirulent".

(15) Line 597- the study by Fukuoka used kidney cells. It is misleading to not clearly state that this finding of ER stress was NOT done in fungi as the way it is presented makes it read as if this work was performed in *C. neoformans*. This should be clarified. This should also be double-checked and clarified for other statements, such as the reference to Harada in line 606, as this study used melanoma cells. These cell types are very different from *cryptococcus*- though I absolutely concur that lessons can be learned from comparative assessments.

We thank the reviewer for pointing out the need to clarify the experimental context of the cited studies. We explicitly stated the host cell types used in the referenced studies by Fukuoka et al. and by Harada et al., respectively, in the revised manuscript (Page 25, lines 560 and 568).

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